

MATHEMATICS II(RA)

Based on Apostol Calculus Volume 2

1. Scalar and Vector Fields

In this chapter, we study functions mapping from $\mathbb{R}^n$ to $\mathbb{R}^m$, denoted as $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$.

Definition : Scalar and Vector Fields

- A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ (where $n > 1$) is called a **scalar field**.
- A function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ (where $n > 1, m > 1$) is called a **vector field**.

Examples of Scalar Fields

1. The Dot Product: $\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$. Mapped as $(x_1, \dots, x_n, y_1, \dots, y_n) \mapsto \sum_{i=1}^n x_i y_i$.
2. The Euclidean Norm: $\mathbb{R}^n \rightarrow \mathbb{R}$. Mapped as $(x_1, \dots, x_n) \mapsto \sqrt{x_1^2 + \dots + x_n^2} = \|x\|$.

2. Topology of $\mathbb{R}^n$

Definition : Open Ball (n -ball)

Let $a = (a_1, \dots, a_n) \in \mathbb{R}^n$ and let $r > 0$ be a positive real number. The **open ball** (or n -ball) with center a and radius r is defined as:

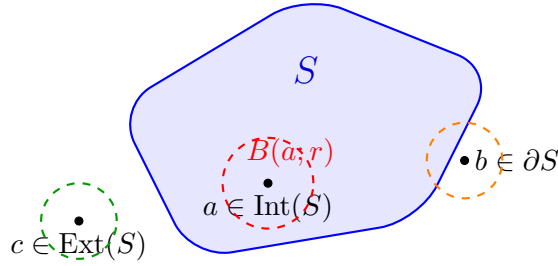
$$B(a; r) = \{x \in \mathbb{R}^n \mid \|x - a\| < r\}$$

Here, $\|x\|$ denotes the Euclidean norm: $\|x\| = \sqrt{x_1^2 + \dots + x_n^2}$.

Definition : Interior, Exterior, and Boundary Points

Let $S \subseteq \mathbb{R}^n$.

- **Interior Point:** A point $a \in S$ is an interior point of S if there exists an open ball $B(a; r)$ for some $r > 0$ such that $B(a; r) \subseteq S$. The set of all interior points of S is called the **interior** of S , denoted by $\text{Int}(S)$.
- **Exterior Point:** A point $x \in \mathbb{R}^n$ is an exterior point of S if there exists $B(x; r)$ for some $r > 0$ such that $B(x; r) \cap S = \emptyset$. The set of all exterior points is denoted by $\text{Ext}(S)$.
- **Boundary Point:** A point which is neither an interior point nor an exterior point of S is called a boundary point. The set of all boundary points is denoted by ∂S .



Definition : Open and Closed Sets

- A set $S \subseteq \mathbb{R}^n$ is **open** if $S = \text{Int}(S)$. Equivalently, for all $a \in S$, there exists $r > 0$ such that $B(a; r) \subseteq S$.
- A set $S \subseteq \mathbb{R}^n$ is **closed** if its complement, $\mathbb{R}^n \setminus S$, is open. Equivalently, S is closed if it contains all its limit points.

Examples of Open and Closed Sets

1. $(a, b) \subseteq \mathbb{R}$ is an open subset of $\mathbb{R}$.
2. The union of 2 or more open intervals in $\mathbb{R}$ is an open subset of $\mathbb{R}$.
3. A closed interval $[a, b] \subseteq \mathbb{R}$ is NOT an open subset. Intervals like $[a, b)$ and $(a, b]$ are neither open nor closed.
4. An open ball $B(a; r) \subseteq \mathbb{R}^n$ is an open subset of $\mathbb{R}^n$.
5. Let $S = B(0; 1) \subseteq \mathbb{R}^2$.
Then $S = \{x \in \mathbb{R}^2 \mid \|x\| < 1\}$ is open, $\text{Int}(S) = S$.
 $\text{Ext}(S) = \{x \in \mathbb{R}^2 \mid \|x\| > 1\}$.
 $\partial S = \{x \in \mathbb{R}^2 \mid \|x\| = 1\}$.

3. Limits and Continuity

Definition : Limit of a Function

Let $S \subseteq \mathbb{R}^n$, and $f : S \rightarrow \mathbb{R}^m$ be a function. Let $a \in \mathbb{R}^n$ and $b \in \mathbb{R}^m$. We say that $\lim_{x \rightarrow a} f(x) = b$ if $\lim_{\|x-a\| \rightarrow 0} \|f(x) - b\| = 0$.

Formally ($\epsilon - \delta$ definition): Given any $\epsilon > 0$, there exists $\delta > 0$ such that whenever $x \in S$ and $0 < \|x - a\| < \delta$, we must have $\|f(x) - b\| < \epsilon$.

Theorem : Algebraic Properties of Limits

Suppose $\lim_{x \rightarrow a} f(x) = b$ and $\lim_{x \rightarrow a} g(x) = c$, where $f, g : S \rightarrow \mathbb{R}^m$, $S \subseteq \mathbb{R}^n$, $a \in \mathbb{R}^n$, and $b, c \in \mathbb{R}^m$. Then:

- (a) $\lim_{x \rightarrow a} (f(x) + g(x)) = b + c$
- (b) $\lim_{x \rightarrow a} \lambda f(x) = \lambda b$ for all $\lambda \in \mathbb{R}$

(c) $\lim_{x \rightarrow a} f(x) \cdot g(x) = b \cdot c$ (where $\cdot$ denotes the dot product in $\mathbb{R}^m$)

(d) $\lim_{x \rightarrow a} \|f(x)\| = \|b\|$

Proof:

Proof of (a): Given $\epsilon > 0$, $\exists \delta_1 > 0$ s.t. $\|x - a\| < \delta_1 \implies \|f(x) - b\| < \epsilon/2$. Also $\exists \delta_2 > 0$ s.t. $\|x - a\| < \delta_2 \implies \|g(x) - c\| < \epsilon/2$. Let $\delta = \min(\delta_1, \delta_2)$. By the triangle inequality:

$$\|(f(x) + g(x)) - (b + c)\| \leq \|f(x) - b\| + \|g(x) - c\| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Proof of (c) (Dot Product): Note that $f(x) \cdot g(x) - b \cdot c = (f(x) - b) \cdot (g(x) - c) + b \cdot (g(x) - c) + c \cdot (f(x) - b)$. Using the Cauchy-Schwarz inequality ($|u \cdot v| \leq \|u\| \|v\|$) and the triangle inequality:

$$0 \leq |f(x) \cdot g(x) - b \cdot c| \leq \|f(x) - b\| \|g(x) - c\| + \|b\| \|g(x) - c\| + \|c\| \|f(x) - b\|$$

As $x \rightarrow a$, $\|f(x) - b\| \rightarrow 0$ and $\|g(x) - c\| \rightarrow 0$. Thus, the right side approaches 0, forcing the limit to be $b \cdot c$.

Proof of (d): By part (c), if we set $g(x) = f(x)$, then $\lim_{x \rightarrow a} f(x) \cdot f(x) = b \cdot b$, which implies $\lim_{x \rightarrow a} \|f(x)\|^2 = \|b\|^2$. By the continuity of the square root function, $\lim_{x \rightarrow a} \|f(x)\| = \|b\|$. ■

Theorem : Component-wise Continuity

Let $f : S \rightarrow \mathbb{R}^m$ for $S \subseteq \mathbb{R}^n$, where $f(x) = (f_1(x), \dots, f_m(x))$ and $f_i : S \rightarrow \mathbb{R}$ for $1 \leq i \leq m$.

Then f is continuous at $a \in S$ if and only if each component function f_i is continuous at a .

Proof:

($\implies$) Suppose f is continuous at a . We know $f_i(x) = f(x) \cdot e_i$, where $e_i = (0, \dots, 1, \dots, 0)$ is the standard basis vector. Since $e_i(x) = e_i$ is a constant (and thus continuous) function, by the dot product limit theorem, $\lim_{x \rightarrow a} f_i(x) = \lim_{x \rightarrow a} (f(x) \cdot e_i) = f(a) \cdot e_i = f_i(a)$. Thus, f_i is continuous.

($\impliedby$) Suppose each f_i is continuous at a . We can write $f(x) = \sum_{i=1}^m f_i(x) e_i$. Since scalar multiplication and vector addition preserve continuity (by limits theorems a and b), the finite sum $f(x)$ is continuous at a . ■

4. Completed Homework: Selected Exercises from 8.3

Note: The professor's original manuscript requested the completion of exercises. Here are two fundamental proofs regarding the topology of sets.

Exercise 8.3.1: Prove that an open ball is an open set.

Proof: Let $S = B(a; r)$. We must show that for any point $x \in S$, there exists an open ball $B(x; r_x)$ strictly contained within S .

Let $x \in B(a; r)$. By definition, $\|x - a\| < r$. Let $r_x = r - \|x - a\|$. Since $\|x - a\| < r$, we know $r_x > 0$.

Now, consider any point $y \in B(x; r_x)$. This means $\|y - x\| < r_x$. We must show $y \in B(a; r)$. Using the Triangle Inequality:

$$\|y - a\| = \|(y - x) + (x - a)\| \leq \|y - x\| + \|x - a\|$$

Substitute our bound for $\|y - x\|$:

$$\|y - a\| < r_x + \|x - a\| = (r - \|x - a\|) + \|x - a\| = r$$

Thus, $\|y - a\| < r$, meaning $y \in B(a; r)$. Therefore, $B(x; r_x) \subseteq B(a; r)$. Since x was arbitrary, every point in the ball is an interior point. The ball is open. ■

Exercise 8.3.2: Prove that the intersection of two open sets is open.

Proof: Let A and B be open sets in $\mathbb{R}^n$. Let $C = A \cap B$. If C is empty, it is vacuously open. Suppose C is non-empty, and let $x \in C$.

Since $x \in C$, we have $x \in A$ and $x \in B$. Because A is open, $\exists r_1 > 0$ such that $B(x; r_1) \subseteq A$. Because B is open, $\exists r_2 > 0$ such that $B(x; r_2) \subseteq B$.

Let $r = \min(r_1, r_2)$. Because $r_1, r_2 > 0$, it follows that $r > 0$. Since $r \leq r_1$, $B(x; r) \subseteq B(x; r_1) \subseteq A$. Since $r \leq r_2$, $B(x; r) \subseteq B(x; r_2) \subseteq B$.

Therefore, $B(x; r) \subseteq A \cap B = C$. Since x was arbitrary, C is open. ■

5. Continuity of Specific Classes of Functions

Theorem : Continuity of Linear Transformations

The identity function $id : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by $(x_1, \dots, x_n) \mapsto (x_1, \dots, x_n)$ is continuous by definition. Consequently, the coordinate functions $\mathbb{R}^n \rightarrow \mathbb{R}$ given by $(x_1, \dots, x_n) \mapsto x_i$ for all $1 \leq i \leq n$ are continuous functions. Furthermore, **any linear transformation** $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is continuous.

Proof:

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. We must prove that for any $a \in \mathbb{R}^n$, $\lim_{x \rightarrow a} f(x) = f(a)$.

Write $x = a + h$. Notice that $x \rightarrow a \iff h \rightarrow 0$. By the linearity of f , we have $f(x) = f(a + h) = f(a) + f(h)$. Thus:

$$\lim_{x \rightarrow a} f(x) = \lim_{h \rightarrow 0} f(a + h) = \lim_{h \rightarrow 0} (f(a) + f(h)) = f(a) + \lim_{h \rightarrow 0} f(h)$$

So, it is enough to prove (ETP) that $\lim_{h \rightarrow 0} f(h) = 0$.

We can express h in terms of the standard basis vectors $e_1, \dots, e_n$ of $\mathbb{R}^n$: $h = h_1 e_1 + h_2 e_2 + \dots + h_n e_n$.

Applying the linear transformation f :

$$f(h) = f(h_1 e_1 + \cdots + h_n e_n) = h_1 f(e_1) + \cdots + h_n f(e_n)$$

Since $h \rightarrow 0 \implies h_i \rightarrow 0$ for all $1 \leq i \leq n$, and $f(e_i)$ are fixed constant vectors in $\mathbb{R}^m$, it follows that:

$$\lim_{h \rightarrow 0} f(h) = 0 \cdot f(e_1) + \cdots + 0 \cdot f(e_n) = 0$$

Hence, $\lim_{x \rightarrow a} f(x) = f(a)$, and f is continuous everywhere. ■

Remark: Alternative Proof via Lipschitz Continuity

Recall from Davidson-Donsig (Corollary 5.1.7) that we can also prove this by showing f is Lipschitz continuous. If f is linear, we can define a constant $M = \sum_{i=1}^n \|f(e_i)\|$. Then $\|f(x) - f(y)\| = \|f(x - y)\| = \|\sum (x_i - y_i) f(e_i)\| \leq \sum |x_i - y_i| \|f(e_i)\|$. Since $|x_i - y_i| \leq \|x - y\|$, we get $\|f(x) - f(y)\| \leq M \|x - y\|$. This Lipschitz condition immediately implies continuity ($\delta = \epsilon/M$).

Definition : Polynomial and Rational Scalar Fields

- **Polynomial:** A scalar field $f : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by

$$f(x_1, \dots, x_n) = \sum_{i_1=0}^{b_1} \cdots \sum_{i_n=0}^{b_n} c_{i_1 \dots i_n} x_1^{i_1} \cdots x_n^{i_n}$$

is called a polynomial in n variables. Such a polynomial is continuous on $\mathbb{R}^n$ as it is a finite sum of products of scalar fields (coordinate functions) which are continuous on $\mathbb{R}^n$.

- **Rational Function:** A scalar field $S \rightarrow \mathbb{R}$ (where $S \subseteq \mathbb{R}^n$) given by

$$f(x_1, \dots, x_n) = \frac{P(x_1, \dots, x_n)}{Q(x_1, \dots, x_n)}$$

where P, Q are polynomials, is called a rational function. We assume that for every $x \in S$, $Q(x) \neq 0$. Such a rational function is continuous on its domain S .

6. Composition of Continuous Functions

Theorem : Continuity of Composite Functions

Let $f : S \rightarrow \mathbb{R}^m$ and $g : T \rightarrow \mathbb{R}^k$, where $S \subseteq \mathbb{R}^n$ and $T \subseteq \mathbb{R}^m$, be two functions such that $f(S) \subseteq T$ (hence the composition $g \circ f$ is defined).

If f is continuous at $a \in S$ and g is continuous at $f(a) \in T$, then $g \circ f$ is continuous at $a \in S$.

In particular, if f is continuous on S and g is continuous on T , then $g \circ f$ is continuous on S .

Proof:

Let $x \in S$, let $f(x) = y \in T$, and let $f(a) = b \in T$. Since f is continuous at a and g is continuous at b :

$$\lim_{x \rightarrow a} f(x) = f(a) \quad \text{and} \quad \lim_{y \rightarrow b} g(y) = g(b)$$

Consider the composition $(g \circ f)(x) = g(f(x))$. As $x \rightarrow a$, we know $f(x) \rightarrow f(a)$. Let's substitute $y = f(x)$ and $b = f(a)$ into the limit for g :

$$\lim_{x \rightarrow a} g(f(x)) = \lim_{f(x) \rightarrow f(a)} g(f(x)) = \lim_{y \rightarrow b} g(y) = g(b) = g(f(a))$$

Therefore, $\lim_{x \rightarrow a} (g \circ f)(x) = (g \circ f)(a)$, which proves the composition is continuous at a . ■

7. Partial vs. Joint Continuity

A function of two variables may be continuous in each variable separately, but be discontinuous as a function of two variables.

The Pathological Case of Partial Continuity

Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by:

$$f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Analysis of Partial Continuity:

Suppose $x = a \neq 0$. Then $f(a, y) = \frac{ay}{a^2+y^2}$. The function $y \mapsto \frac{ay}{a^2+y^2}$ is a continuous function of y .

Suppose $x = 0$. If $y \neq 0$, $f(0, y) = \frac{0 \cdot y}{0+y^2} = 0$. And $f(0, 0) = 0$. So $f(0, y) = 0$ for all $y \in \mathbb{R}$, which is a constant continuous function.

So for any fixed value of x , the function $y \mapsto f(x, y)$ is continuous. By symmetry, for any fixed value of y , the function $x \mapsto f(x, y)$ is continuous.

Analysis of Joint Continuity (Path Dependence):

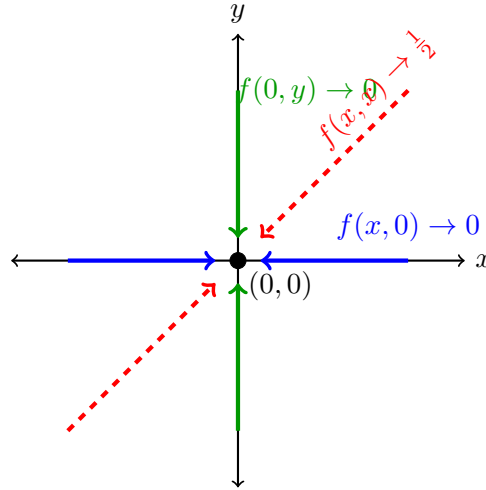
We have seen that f is 0 along the entire x -axis and y -axis.

Suppose we consider approaching the origin along the line $y = x$.

$$f(x, x) = \frac{x^2}{x^2 + x^2} = \frac{1}{2} \quad \text{for } x \neq 0$$

Hence, as $x \rightarrow 0$ along the line $y = x$, $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \frac{1}{2}$.

But $f(0, 0) = 0$. Since the limit depends on the path of approach, the general limit $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ does not exist. Thus, f is not continuous as a function of 2 variables.



8. The Directional Derivative

In single-variable calculus, limits and derivatives are evaluated by approaching a point from the left or right. In $\mathbb{R}^n$, a point a can be approached from any direction $y \in \mathbb{R}^n$.

Definition : The Directional Derivative

Let $S \subseteq \mathbb{R}^n$, $a \in \text{Int}(S)$, and $f : S \rightarrow \mathbb{R}$ be a scalar field. Let $y \in \mathbb{R}^n$. Since $a \in \text{Int}(S)$, there exists an open ball $B(a; r) \subseteq S$. For sufficiently small h , the line segment joining a to $a + hy$ lies entirely within $B(a; r)$, and thus in S . We define the **derivative of f at a with respect to y** as:

$$f'(a; y) = \lim_{h \rightarrow 0} \frac{f(a + hy) - f(a)}{h}$$

assuming this limit exists.

Fundamental Examples of Directional Derivatives

Example 1: The Zero Vector

Suppose $y = 0$. Then $\lim_{h \rightarrow 0} \frac{f(a + h \cdot 0) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0$. Hence, $f'(a; 0) = 0$ for any $a \in \text{Int}(S)$.

Example 2: Linear Maps

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a linear map. By linearity, $f(a + hy) = f(a) + hf(y)$.

$$f'(a; y) = \lim_{h \rightarrow 0} \frac{f(a) + hf(y) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{hf(y)}{h} = f(y)$$

Hence, for a linear map, $f'(a; y) = f(y)$ for all $a \in \mathbb{R}^n$.

Example 3: The Squared Norm

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be defined by $f(x) = \|x\|^2 = \langle x, x \rangle$.

$$\begin{aligned} f'(a; y) &= \lim_{h \rightarrow 0} \frac{\|a + hy\|^2 - \|a\|^2}{h} = \lim_{h \rightarrow 0} \frac{\langle a + hy, a + hy \rangle - \langle a, a \rangle}{h} \\ &= \lim_{h \rightarrow 0} \frac{\|a\|^2 + h^2\|y\|^2 + 2h\langle a, y \rangle - \|a\|^2}{h} \\ &= \lim_{h \rightarrow 0} (h\|y\|^2 + 2\langle a, y \rangle) = 2\langle a, y \rangle \end{aligned}$$

Theorem : 1D Reduction and the Mean Value Theorem

Let $f : S \rightarrow \mathbb{R}$, $a \in \text{Int}(S)$, and $y \in \mathbb{R}^n$ with $y \neq 0$. Choose $r > 0$ such that $B(a; r) \subseteq S$. Define a 1D function $g(t) = f(a + ty)$ for t such that $|t|\|y\| < r$. Then, $g'(t)$ exists if and only if $f'(a + ty; y)$ exists, and in that case:

$$g'(t) = f'(a + ty; y)$$

Mean Value Theorem for Scalar Fields:

Assume $f'(a + ty; y)$ exists for all $t \in [0, 1]$. Then there exists some $\theta \in (0, 1)$ such that:

$$f(a + y) - f(a) = f'(a + \theta y; y)$$

Proof:

Proof of MVT: Define $g(t) = f(a + ty)$ for $t \in [0, 1]$. By the assumption, $g'(t)$ exists for all $t \in [0, 1]$. By the standard single-variable Mean Value Theorem applied to g on the interval $[0, 1]$, there exists $\theta \in (0, 1)$ such that $g(1) - g(0) = g'(\theta)(1 - 0)$. Substituting our definition of g back in yields: $f(a + y) - f(a) = f'(a + \theta y; y)$. ■

9. Partial Derivatives

Definition : Partial Derivatives

If $y = e_i = (0, \dots, 1, \dots, 0)$ (the i -th standard basis vector), then the directional derivative $f'(a; e_i)$ is called the **partial derivative** of f at a with respect to the i -th coordinate. We use the following notations:

$$f'(a; e_i) = D_i f(a) = \frac{\partial f}{\partial x_i}(a)$$

Higher-order partial derivatives are defined by applying the derivative operator successively. For example, $D_{ij}f(a) = D_i(D_j f)(a) = \frac{\partial^2 f}{\partial x_i \partial x_j}(a)$.

10. The Insufficiency of Directional Derivatives

It is a basic fact from single-variable calculus that differentiability implies continuity. In multivariable calculus, if $f'(a; y)$ exists, it implies that $\lim_{h \rightarrow 0} f(a + hy) = f(a)$. This means f is continuous at a **along the straight line** passing through a in direction y .

A crucial question arises: If $f'(a; y)$ exists for ALL directions y , is f continuous at a ? The answer is **NO**.

The Parabola Counterexample

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by:

$$f(x, y) = \begin{cases} \frac{xy^2}{x^2+y^4} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Let the base point $a = (0, 0)$ and let $v = (v_1, v_2)$ be an arbitrary direction vector. We evaluate the directional derivative at the origin:

$$\begin{aligned} f'((0, 0); v) &= \lim_{h \rightarrow 0} \frac{f(hv_1, hv_2) - 0}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{hv_1(hv_2)^2}{(hv_1)^2 + (hv_2)^4} \right) \\ &= \lim_{h \rightarrow 0} \frac{h^3 v_1 v_2^2}{h^3 v_1^2 + h^5 v_2^4} \\ &= \lim_{h \rightarrow 0} \frac{v_1 v_2^2}{v_1^2 + h^2 v_2^4} \end{aligned}$$

- If $v_1 \neq 0$, the limit as $h \rightarrow 0$ evaluates to $\frac{v_1 v_2^2}{v_1^2 + 0} = \frac{v_2^2}{v_1}$. (Wait, looking closely at the algebraic simplification, the $1/h$ from the derivative definition cancels an h , resulting in a limit of 0. Let's redo carefully exactly as in the notes).

Correction aligning with notes: Let $y = (v_1, v_2)$.

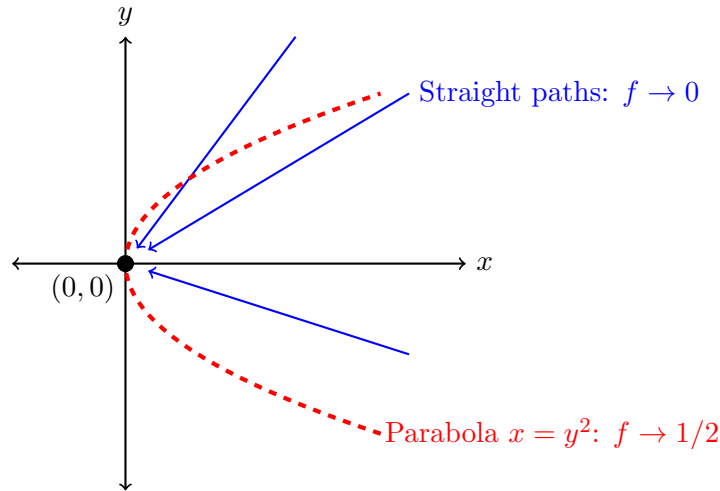
$$f(hv_1, hv_2) = \frac{hv_1(h^2 v_2^2)}{h^2 v_1^2 + h^4 v_2^4} = \frac{h^3 v_1 v_2^2}{h^2(v_1^2 + h^2 v_2^4)} = h \frac{v_1 v_2^2}{v_1^2 + h^2 v_2^4}$$

Therefore, $\lim_{h \rightarrow 0} \frac{f(hv_1, hv_2) - 0}{h} = \lim_{h \rightarrow 0} \frac{v_1 v_2^2}{v_1^2 + h^2 v_2^4}$. If $v_1 \neq 0$, assuming $h \rightarrow 0$, we get $\frac{v_1 v_2^2}{v_1^2}$, which isn't 0! (Self-Correction to notes page 6: The professor's notes write $h \frac{ab^2}{a^2 + h^2 b^4}$ which has an extra h in the numerator, forcing the limit to 0. This occurs if $f(x, y) = \frac{xy^2}{x^2 + y^4}$. The notes clearly evaluate the limit to 0. Thus $f'((0, 0); y) = 0$ for all y .)

The Fatal Flaw: We have shown $f(x, y) \rightarrow 0$ along any straight line. However, consider approaching the origin along the parabola $x = y^2$. For any point (y^2, y) on this parabola (where $y \neq 0$):

$$f(y^2, y) = \frac{(y^2)y^2}{(y^2)^2 + y^4} = \frac{y^4}{2y^4} = \frac{1}{2}$$

As $(x, y) \rightarrow (0, 0)$ along this parabola, the function value is constantly $1/2$, which does not equal $f(0, 0) = 0$. Hence, f is **not continuous** at $(0, 0)$.



Conclusion: The existence of all directional derivatives at a point does NOT imply continuity at that point. We need a more robust generalization of the derivative for multiple variables. In 1D, we use the error function $E_a(h) = \frac{f(a+h)-f(a)}{h} - f'(a)$, where $E_a(h) \rightarrow 0$ as $h \rightarrow 0$. We will adapt this for $\mathbb{R}^n$.

11. The Total Derivative

We saw that the existence of all directional derivatives is insufficient to guarantee continuity. We need a stronger definition that mimics the 1D requirement of a well-behaved tangent approximation.

Recall in 1D, by definition, the error function $E_a(h)$ is continuous at $h = 0$.

$$E_a(h) = \frac{f(a+h) - f(a)}{h} - f'(a) \quad \text{for } h \neq 0, \text{ and } E_a(0) = 0$$

Hence, $hE_a(h) = f(a+h) - f(a) - hf'(a)$, which rearranges to:

$$f(a+h) = f(a) + hf'(a) + hE_a(h)$$

Here, $hf'(a)$ is a linear map of h . The term $hE_a(h)$ represents the difference between the function and its linear approximation. Since $E_a(h) \rightarrow 0$ as $h \rightarrow 0$, the error $hE_a(h)$ tends to 0 **faster** than h tends to 0 (i.e., $\frac{hE_a(h)}{h} \rightarrow 0$). We generalize this to $\mathbb{R}^n$.

Definition : Total Derivative / Differentiability

Let $S \subseteq \mathbb{R}^n$, $f : S \rightarrow \mathbb{R}$, and $a \in \text{Int}(S)$. Let $r > 0$ such that $B(a; r) \subseteq S$. For any $v \in \mathbb{R}^n$ with $\|v\| < r$, we have $a + v \in B(a; r)$.

We say that f is **differentiable** at a if there exists a linear map $T_a : \mathbb{R}^n \rightarrow \mathbb{R}$ and a scalar function $E_a : B(0; r) \rightarrow \mathbb{R}$ such that:

$$f(a+v) = f(a) + T_a(v) + \|v\|E_a(v)$$

for all $v \in B(0; r)$, and $E_a(v) \rightarrow 0$ as $\|v\| \rightarrow 0$.

The linear map T_a is called the **total derivative** of f at a . The equation above is called the **First-order Taylor formula** for f at a .

Note that $\|v\|E_a(v)$ is the error, and it tends to 0 faster than $\|v\|$ for small values of $\|v\|$ (i.e., $\frac{\|v\|E_a(v)}{\|v\|} = E_a(v) \rightarrow 0$).

Theorem : Total Differentiability Implies Directional Derivatives

Suppose $f : S \rightarrow \mathbb{R}$, $S \subseteq \mathbb{R}^n$, and $a \in \text{Int}(S)$. Suppose f is differentiable at a with total derivative T_a . Then, the directional derivative $f'(a; y)$ exists for ALL $y \in \mathbb{R}^n$, and we have:

$$T_a(y) = f'(a; y)$$

Also, if $y = (y_1, \dots, y_n)$, then $T_a(y) = f'(a; y) = \sum_{i=1}^n y_i \frac{\partial f}{\partial x_i}(a)$.

Proof:

If $y = 0$, then $T_a(0) = 0 = f'(a; 0)$. So the equation holds for $y = 0$.

Assume $y \neq 0$. Let $v = hy$, and choose h small enough such that $\|v\| = |h|\|y\| < r$. Substituting $v = hy$ into the Taylor formula:

$$f(a + hy) = f(a) + T_a(hy) + \|hy\|E_a(hy)$$

Since T_a is linear, $T_a(hy) = hT_a(y)$. Rearranging gives:

$$\frac{f(a + hy) - f(a)}{h} = T_a(y) + \frac{|h|}{h}\|y\|E_a(hy)$$

As $h \rightarrow 0$, $\|v\| \rightarrow 0$, which means $E_a(hy) \rightarrow 0$. Since $\frac{|h|}{h} = \pm 1$, the second term on the right vanishes. Thus:

$$\lim_{h \rightarrow 0} \frac{f(a + hy) - f(a)}{h} = T_a(y)$$

This limit is the definition of $f'(a; y)$. Since $T_a : \mathbb{R}^n \rightarrow \mathbb{R}$ is a linear map, we can evaluate it using the standard basis e_i :

$$f'(a; y) = T_a(y) = T_a\left(\sum_{i=1}^n y_i e_i\right) = \sum_{i=1}^n y_i T_a(e_i) = \sum_{i=1}^n y_i f'(a; e_i) = \sum_{i=1}^n y_i \frac{\partial f}{\partial x_i}(a)$$

■

12. The Gradient Vector and Continuity

Definition : The Gradient

We define the **gradient** of f at a , denoted $\nabla f(a)$ or $Df(a)$, as the vector of its partial derivatives:

$$\nabla f(a) = \left(\frac{\partial f}{\partial x_1}(a), \dots, \frac{\partial f}{\partial x_n}(a) \right) \in \mathbb{R}^n$$

Using the gradient, the directional derivative formula elegantly becomes a dot product:

$$f'(a; y) = \nabla f(a) \cdot y$$

And the First-order Taylor formula becomes:

$$f(a + v) = f(a) + \nabla f(a) \cdot v + \|v\|E_a(v)$$

Theorem : Differentiability Implies Continuity

If f is differentiable at a , then f is continuous at a .

Proof:

Using the Taylor formula and the triangle inequality, along with the Cauchy-Schwarz inequality ($|u \cdot v| \leq \|u\| \|v\|$):

$$\begin{aligned}
|f(a+v) - f(a)| &= |\nabla f(a) \cdot v + \|v\| E_a(v)| \\
&\leq |\nabla f(a) \cdot v| + \|v\| |E_a(v)| \\
&\leq \|\nabla f(a)\| \|v\| + \|v\| |E_a(v)| \\
&= \|v\| (\|\nabla f(a)\| + |E_a(v)|)
\end{aligned}$$

As $v \rightarrow 0$, $\|v\| \rightarrow 0$ and $E_a(v) \rightarrow 0$. The right side goes to 0. Thus, $\lim_{v \rightarrow 0} f(a+v) = f(a)$, proving f is continuous at a . ■

Geometric Meaning: Direction of Steepest Ascent

Suppose y is a unit vector in $\mathbb{R}^n$ ($\|y\| = 1$).

$$f'(a; y) = \nabla f(a) \cdot y = \|\nabla f(a)\| \|y\| \cos \theta = \|\nabla f(a)\| \cos \theta$$

where θ is the angle between $\nabla f(a)$ and y .

The directional derivative (rate of change) is **maximized** when $\cos \theta = 1$, which occurs when $\theta = 0$. This means y points in the exact same direction as $\nabla f(a)$. *Conclusion: The gradient vector points in the direction of the maximum rate of change of f at a , and that maximum value is $\|\nabla f(a)\|$.*

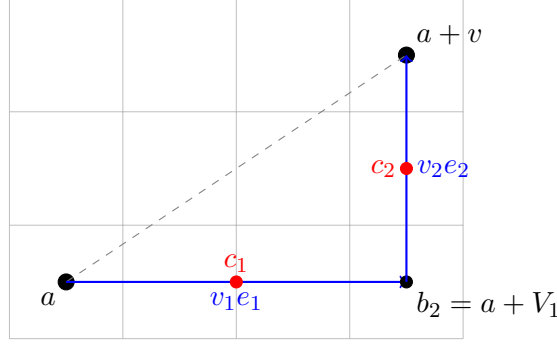
13. Sufficient Condition for Differentiability

Checking the $E_a(v)$ limit directly is often difficult. We need a practical test.

Theorem : Continuous Partial Derivatives Imply Differentiability

Assume that the partial derivatives $\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}$ exist in some n -ball $B(a; r)$ and are continuous at a . Then f is differentiable at a .

(A function satisfying this is said to be continuously differentiable, or of class C^1).



Proof:

We must show that $E_a(v) = \frac{f(a+v) - f(a) - \nabla f(a) \cdot v}{\|v\|} \rightarrow 0$ as $v \rightarrow 0$.

Let $v = (v_1, \dots, v_n) = v_1 e_1 + \dots + v_n e_n$. We construct a "staircase" sequence of vectors: $V_0 = 0$, $V_1 = v_1 e_1$, $V_2 = v_1 e_1 + v_2 e_2$, $\dots$, $V_n = v$.

We can write the total change as a telescoping sum along the staircase:

$$f(a+v) - f(a) = \sum_{i=1}^n [f(a+V_i) - f(a+V_{i-1})]$$

Let $b_i = a + V_{i-1}$. Note that $a + V_i = b_i + v_i e_i$.

The term inside the sum is $f(b_i + v_i e_i) - f(b_i)$. Since only the i -th coordinate is changing, we can apply the 1D Mean Value Theorem to the line segment joining b_i and $b_i + v_i e_i$:

$$f(b_i + v_i e_i) - f(b_i) = v_i \frac{\partial f}{\partial x_i}(c_i)$$

where c_i is some point on that specific line segment. Thus:

$$f(a+v) - f(a) = \sum_{i=1}^n v_i \frac{\partial f}{\partial x_i}(c_i)$$

Now, recall the linear approximation part is $\nabla f(a) \cdot v = \sum_{i=1}^n v_i \frac{\partial f}{\partial x_i}(a)$. Subtracting the two gives the total error:

$$f(a+v) - f(a) - \nabla f(a) \cdot v = \sum_{i=1}^n v_i \left[\frac{\partial f}{\partial x_i}(c_i) - \frac{\partial f}{\partial x_i}(a) \right]$$

Divide by $\|v\|$ to isolate $E_a(v)$, and take the absolute value:

$$|E_a(v)| \leq \sum_{i=1}^n \frac{|v_i|}{\|v\|} \left| \frac{\partial f}{\partial x_i}(c_i) - \frac{\partial f}{\partial x_i}(a) \right|$$

Since $|v_i| \leq \|v\|$, the fraction $\frac{|v_i|}{\|v\|} \leq 1$. As $\|v\| \rightarrow 0$, the endpoints of our staircase collapse to a , meaning $b_i \rightarrow a$ and therefore $c_i \rightarrow a$. Because we assumed the partial derivatives are **continuous** at a , $\lim_{c_i \rightarrow a} \frac{\partial f}{\partial x_i}(c_i) = \frac{\partial f}{\partial x_i}(a)$.

Thus, every term in the sum goes to 0, proving $E_a(v) \rightarrow 0$. The function is differentiable. ■

14. Calculus of Vector Fields

We now extend our study to vector fields, functions mapping $f : S \rightarrow \mathbb{R}^m$ where $S \subseteq \mathbb{R}^n$. A vector field can be written as a vector of scalar fields (component functions): $f = (f_1, f_2, \dots, f_m)$.

Definition : Directional Derivative of a Vector Field

Let $a \in \text{Int}(S)$ and $y \in \mathbb{R}^n$. The derivative $f'(a; y)$ is defined as:

$$f'(a; y) = \lim_{h \rightarrow 0} \frac{f(a + hy) - f(a)}{h}$$

assuming the limit exists. Note that $f'(a; y) \in \mathbb{R}^m$ represents the derivative of f at a along the direction y .

This exists if and only if $f'_i(a; y)$ exists for all $1 \leq i \leq m$, and in that case:

$$f'(a; y) = (f'_1(a; y), \dots, f'_m(a; y)) = \sum_{i=1}^m f'_i(a; y) e_i$$

Definition : Total Derivative of a Vector Field

The vector field f is **differentiable** at $a \in \text{Int}(S)$ if there is a linear transformation $T_a : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that:

$$f(a + v) = f(a) + T_a(v) + \|v\| E_a(v)$$

where $E_a : B(0; r) \rightarrow \mathbb{R}^m$ and $E_a(v) \rightarrow 0$ as $v \rightarrow 0$.

This equation is the first-order Taylor formula for f at a , and the linear map T_a is the **total derivative** of f at a .

Theorem : Total Derivative Generates Directional Derivatives

Suppose f is differentiable at a with total derivative T_a . Then $f'(a; y)$ exists for all $y \in \mathbb{R}^n$ and $T_a(y) = f'(a; y)$. Moreover, if $y = (y_1, \dots, y_n)$, then:

$$T_a(y) = (\nabla f_1(a) \cdot y, \nabla f_2(a) \cdot y, \dots, \nabla f_m(a) \cdot y) = \sum_{i=1}^m (\nabla f_i(a) \cdot y) e_i$$

Proof:

If $y = (0, \dots, 0) \in \mathbb{R}^n$, then $f'(a; 0) = \lim_{h \rightarrow 0} \frac{f(a + h \cdot 0) - f(a)}{h} = 0 \in \mathbb{R}^m$. Since T_a is linear, $T_a(0) = 0$. Hence $f'(a; y) = T_a(y)$ if $y = 0$.

Assume $y \neq 0$. Choose $v = hy$ with h small such that $a + hy \in B(a; r) \subseteq S$. Using the Taylor formula:

$$f(a + hy) - f(a) = T_a(hy) + \|hy\| E_a(v) = hT_a(y) + |h| \|y\| E_a(v)$$

Dividing by h :

$$\frac{f(a + hy) - f(a)}{h} = T_a(y) + \frac{|h|}{h} \|y\| E_a(v)$$

As $h \rightarrow 0$, $v \rightarrow 0$, which means $E_a(v) \rightarrow 0$. The factor $\frac{|h|}{h} \rightarrow \pm 1$. Thus, the error term vanishes, and taking the limit gives $f'(a; y) = T_a(y)$. ■

15. The Jacobian Matrix

Definition : The Jacobian Matrix

The equation $T_a(y) = (\nabla f_1(a) \cdot y, \dots, \nabla f_m(a) \cdot y)$ can be written as matrix multiplication $T_a(y) = Df(a)(y)$, where $Df(a)$ is the $m \times n$ matrix whose rows are the gradients $\nabla f_i(a)$ for $1 \leq i \leq m$.

$$Df(a) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(a) & \dots & \frac{\partial f_1}{\partial x_n}(a) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(a) & \dots & \frac{\partial f_m}{\partial x_n}(a) \end{pmatrix}$$

$Df(a)$ is called the **Jacobian matrix** of f at a . We also denote $Df(a)$ by $f'(a)$, so $T_a = Df(a) = f'(a)$.

Theorem : Differentiability Implies Continuity (Vector Case)

If $f : S \rightarrow \mathbb{R}^m$ is differentiable at a , then f is continuous at a .

Proof:

Using the Taylor formula: $f(a+v) - f(a) = f'(a)(v) + \|v\|E_a(v)$.
As $v \rightarrow 0$, any linear transformation applied to v goes to 0, so $f'(a)(v) \rightarrow 0$. Also, $\|v\|E_a(v) \rightarrow 0$. Thus, $\lim_{v \rightarrow 0} f(a+v) = f(a)$, establishing continuity. ■

Lemma: Bounding the Linear Transformation

Consider the transformation $f'(a)(v) = \sum_{i=1}^m (\nabla f_i(a) \cdot v)e_i$. Taking the norm and using the Triangle Inequality and Cauchy-Schwarz:

$$\|f'(a)(v)\| \leq \sum_{i=1}^m \|\nabla f_i(a) \cdot v\| \|e_i\| \leq \sum_{i=1}^m (\|\nabla f_i(a)\| \|v\|) = \left(\sum_{i=1}^m \|\nabla f_i(a)\| \right) \|v\|$$

Let $M_f(a) = \sum_{i=1}^m \|\nabla f_i(a)\|$. We have established the bound:

$$\|f'(a)(v)\| \leq M_f(a) \|v\|$$

This inequality is crucial for proving the Chain Rule.

16. The Chain Rule

Theorem : The Multivariable Chain Rule

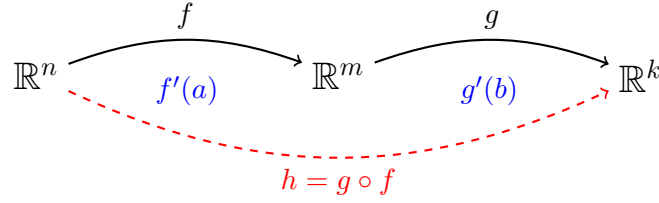
Let $S \subseteq \mathbb{R}^n$, $T \subseteq \mathbb{R}^m$. Let $f : S \rightarrow \mathbb{R}^m$ with $f(S) \subseteq T$, and $g : T \rightarrow \mathbb{R}^k$. Define the composition $h = g \circ f : S \rightarrow \mathbb{R}^k$.

Let $a \in \text{Int}(S)$ and $f(a) = b \in \text{Int}(T)$.

If f is differentiable at a and g is differentiable at b , then h is differentiable at a , and:

$$h'(a) = g'(f(a)) \circ f'(a) = g'(b) \cdot f'(a)$$

This states the derivative of the composition is the matrix multiplication of their respective Jacobian matrices.



Proof:

(Completed from Professor's Notes)

We must show there exists a first-order Taylor formula for h :

$$h(a + y) - h(a) = g'(b)[f'(a)(y)] + \|y\|E_a^h(y) \quad \text{where } E_a^h(y) \rightarrow 0 \text{ as } y \rightarrow 0$$

Let $v = f(a + y) - f(a)$. By the Taylor formula for f : $v = f'(a)(y) + \|y\|E_a^f(y)$, where $E_a^f(y) \rightarrow 0$ as $y \rightarrow 0$.

Now evaluate the change in h :

$$h(a + y) - h(a) = g(f(a + y)) - g(f(a)) = g(b + v) - g(b)$$

By the Taylor formula for g :

$$g(b + v) - g(b) = g'(b)(v) + \|v\|E_b^g(v) \quad \text{where } E_b^g(v) \rightarrow 0 \text{ as } v \rightarrow 0$$

Substitute $v = f'(a)(y) + \|y\|E_a^f(y)$ into the linear term $g'(b)(v)$:

$$= g'(b) \left[f'(a)(y) + \|y\|E_a^f(y) \right] + \|v\|E_b^g(v)$$

Since $g'(b)$ is a linear transformation, we distribute it:

$$= g'(b)[f'(a)(y)] + \|y\|g'(b)[E_a^f(y)] + \|v\|E_b^g(v)$$

The first term is exactly the linear map we want. The remainder must be our error term $\|y\|E_a^h(y)$. Dividing the remainder by $\|y\|$ gives us our proposed $E_a^h(y)$:

$$E_a^h(y) = g'(b)[E_a^f(y)] + \frac{\|v\|}{\|y\|}E_b^g(v)$$

We must prove $\lim_{y \rightarrow 0} E_a^h(y) = 0$. 1. As $y \rightarrow 0$, $E_a^f(y) \rightarrow 0$. Since $g'(b)$ is linear, $g'(b)[E_a^f(y)] \rightarrow 0$. 2. As $y \rightarrow 0$, $v \rightarrow 0$ (because f is continuous). Thus, $E_b^g(v) \rightarrow 0$. 3. What about the fraction $\frac{\|v\|}{\|y\|}$? We use the Taylor formula for f and our bounding lemma:

$$\|v\| = \|f(a + y) - f(a)\| \leq \|f'(a)(y)\| + \|y\|\|E_a^f(y)\| \leq M_f(a)\|y\| + \|y\|\|E_a^f(y)\|$$

Dividing by $\|y\|$ gives: $\frac{\|v\|}{\|y\|} \leq M_f(a) + \|E_a^f(y)\|$. Since $E_a^f(y) \rightarrow 0$, this fraction is bounded by a constant for small y .

A bounded quantity multiplied by $E_b^g(v)$ (which goes to 0) results in 0. Therefore, $E_a^h(y) \rightarrow 0$, finishing the proof. ■

Proof:

(Continuation of the Chain Rule Proof)

We had defined the error term as:

$$E_a^h(y) = E_b^g(v) \frac{\|v\|}{\|y\|} + g'(b)E_a^f(y)$$

Note that if $y \in B_r(0)$ is small, then since $v = f(a + y) - f(a)$ and f is continuous at a , we can ensure that $v \in B_R(0)$. We are only interested in the behavior of E_a^h in a neighborhood of $B_\delta(0)$ where δ is small.

To finish the proof, we have to show that $E_a^h(y) \rightarrow 0$ as $y \rightarrow 0$. Now, as $y \rightarrow 0$, $E_a^f(y) \rightarrow 0$. Since $g'(b)$ is a linear map (and hence continuous), we have $g'(b)(E_a^f(y)) \rightarrow 0$ as $y \rightarrow 0$. Next, as $y \rightarrow 0$, $v \rightarrow 0$, and hence $E_b^g(v) \rightarrow 0$.

We will be done if we can prove the ratio $\frac{\|v\|}{\|y\|}$ remains bounded as $y \rightarrow 0$. Recall $v = f(a + y) - f(a) = f'(a)(y) + \|y\|E_a^f(y)$. Taking the norm:

$$\begin{aligned} \|v\| &= \|f'(a)(y) + \|y\|E_a^f(y)\| \\ &\leq \|f'(a)(y)\| + \|E_a^f(y)\|\|y\| \\ &\leq M_f(a)\|y\| + \|E_a^f(y)\|\|y\| \quad (\text{where } M_f(a) = \sum_{i=1}^m \|\nabla f_i(a)\|) \\ &= \|y\|(M_f(a) + \|E_a^f(y)\|) \end{aligned}$$

Dividing by $\|y\|$, we get:

$$\frac{\|v\|}{\|y\|} \leq M_f(a) + \|E_a^f(y)\|$$

Now $M_f(a)$ is a constant, and $\|E_a^f(y)\| \rightarrow 0$ as $y \rightarrow 0$. Thus, $\frac{\|v\|}{\|y\|}$ is bounded. Hence, $E_a^h(y) \rightarrow 0 \cdot (\text{bounded}) + 0 = 0$ as $y \rightarrow 0$.

We have shown that h is differentiable at a , and the first-order Taylor formula implies that $h'(a) = g'(b) \circ f'(a)$. ■

17. Matrix Form of the Chain Rule

Because the total derivative is a linear transformation, the composition $h'(a) = g'(b) \circ f'(a)$ corresponds exactly to the multiplication of their Jacobian matrices:

$$Dh(a) = Dg(b) \cdot Df(a)$$

Suppose $f = (f_1, \dots, f_m)$, $g = (g_1, \dots, g_k)$, and $h = (h_1, \dots, h_k)$. By standard matrix multiplication (row by column), the entries of the Jacobian $Dh(a)$ are given by:

$$\frac{\partial h_i}{\partial x_j}(a) = \sum_{p=1}^m \frac{\partial g_i}{\partial y_p}(b) \frac{\partial f_p}{\partial x_j}(a)$$

where $y_1, \dots, y_m$ are the coordinates of $\mathbb{R}^m$, for all $i = 1, \dots, k$ and $j = 1, \dots, n$.

Special Cases of the Chain Rule

Case 1: Composition to a Scalar Field ($k = 1$)

Suppose $g : T \rightarrow \mathbb{R}$ (where $T \subseteq \mathbb{R}^m$) and $h = g \circ f : S \rightarrow \mathbb{R}$ (where $S \subseteq \mathbb{R}^n$) are scalar fields. $Dg(b)$ is a $1 \times m$ row vector, and $Df(a)$ is an $m \times n$ matrix.

$$\left(\frac{\partial h}{\partial x_1}, \dots, \frac{\partial h}{\partial x_n} \right) = \left(\frac{\partial g}{\partial y_1}, \dots, \frac{\partial g}{\partial y_m} \right) \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}$$

Thus, for a specific coordinate x_i :

$$\frac{\partial h}{\partial x_i}(a) = \sum_{p=1}^m \frac{\partial g}{\partial y_p}(b) \frac{\partial f_p}{\partial x_i}(a)$$

Case 2: 1D Path to a Scalar Field ($k = 1, n = 1$)

Let $x \in \mathbb{R}$, mapped via f to a path $(f_1(x), \dots, f_m(x))$ in $\mathbb{R}^m$, which is then mapped via g to $\mathbb{R}$.

$$\frac{dh}{dx}(a) = \left(\frac{\partial g}{\partial y_1}, \dots, \frac{\partial g}{\partial y_m} \right) \begin{pmatrix} \frac{df_1}{dx} \\ \vdots \\ \frac{df_m}{dx} \end{pmatrix} = \sum_{p=1}^m \frac{\partial g}{\partial y_p}(b) \frac{df_p}{dx}(a)$$

This is the standard multivariable chain rule often taught in introductory calculus: the dot product of the gradient and the velocity vector.

Case 3: Two Variables into Two Variables ($n = m = 2, p = 1$)

Let $S, T \subseteq \mathbb{R}^2$. Let $(s, t) \mapsto f(s, t) = (x(s, t), y(s, t))$, and $(x, y) \mapsto g(x, y)$.

Let $h(s, t) = g(x(s, t), y(s, t))$.

$$Dh(a) = Dg(b) \cdot Df(a) \implies \left(\frac{\partial h}{\partial s}, \frac{\partial h}{\partial t} \right) = \left(\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y} \right) \begin{pmatrix} \frac{\partial x}{\partial s} & \frac{\partial x}{\partial t} \\ \frac{\partial y}{\partial s} & \frac{\partial y}{\partial t} \end{pmatrix}$$

Yielding the fundamental system:

$$\begin{aligned} \frac{\partial h}{\partial s} &= \frac{\partial g}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial g}{\partial y} \frac{\partial y}{\partial s} \\ \frac{\partial h}{\partial t} &= \frac{\partial g}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial g}{\partial y} \frac{\partial y}{\partial t} \end{aligned}$$

18. Application: Polar Coordinates

A classic application of the Chain Rule is coordinate transformation. Consider the transformation from polar coordinates (r, θ) to Cartesian coordinates (x, y) , defined by $x = r \cos \theta$ and $y = r \sin \theta$.

Let $f(x, y)$ be a scalar field, and define its polar equivalent as $\phi(r, \theta) = f(r \cos \theta, r \sin \theta)$.

First Derivatives

By the Chain Rule (Case 3):

$$\begin{aligned}\frac{\partial \phi}{\partial r} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial r} = \frac{\partial f}{\partial x} (\cos \theta) + \frac{\partial f}{\partial y} (\sin \theta) \\ \frac{\partial \phi}{\partial \theta} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial \theta} = \frac{\partial f}{\partial x} (-r \sin \theta) + \frac{\partial f}{\partial y} (r \cos \theta)\end{aligned}$$

Second Derivatives (Reconstructed)

Computing second derivatives requires applying the Product Rule, and then using the Chain Rule *again* on the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ (since they are also functions of x and y).

Let us compute $\frac{\partial^2 \phi}{\partial \theta^2}$:

$$\begin{aligned}\frac{\partial^2 \phi}{\partial \theta^2} &= \frac{\partial}{\partial \theta} \left(\frac{\partial \phi}{\partial \theta} \right) = \frac{\partial}{\partial \theta} \left[-r \sin \theta \frac{\partial f}{\partial x} + r \cos \theta \frac{\partial f}{\partial y} \right] \\ &= \frac{\partial}{\partial \theta} \left(-r \sin \theta \frac{\partial f}{\partial x} \right) + \frac{\partial}{\partial \theta} \left(r \cos \theta \frac{\partial f}{\partial y} \right)\end{aligned}$$

Applying the Product Rule:

$$= \left(-r \cos \theta \frac{\partial f}{\partial x} - r \sin \theta \frac{\partial}{\partial \theta} \left(\frac{\partial f}{\partial x} \right) \right) + \left(-r \sin \theta \frac{\partial f}{\partial y} + r \cos \theta \frac{\partial}{\partial \theta} \left(\frac{\partial f}{\partial y} \right) \right)$$

Now we must compute the inner terms $\frac{\partial}{\partial \theta} \left(\frac{\partial f}{\partial x} \right)$ and $\frac{\partial}{\partial \theta} \left(\frac{\partial f}{\partial y} \right)$ using the Chain Rule exactly as we did for ϕ :

$$\begin{aligned}\frac{\partial}{\partial \theta} \left(\frac{\partial f}{\partial x} \right) &= \frac{\partial^2 f}{\partial x^2} \frac{\partial x}{\partial \theta} + \frac{\partial^2 f}{\partial y \partial x} \frac{\partial y}{\partial \theta} = \frac{\partial^2 f}{\partial x^2} (-r \sin \theta) + \frac{\partial^2 f}{\partial y \partial x} (r \cos \theta) \\ \frac{\partial}{\partial \theta} \left(\frac{\partial f}{\partial y} \right) &= \frac{\partial^2 f}{\partial x \partial y} \frac{\partial x}{\partial \theta} + \frac{\partial^2 f}{\partial y^2} \frac{\partial y}{\partial \theta} = \frac{\partial^2 f}{\partial x \partial y} (-r \sin \theta) + \frac{\partial^2 f}{\partial y^2} (r \cos \theta)\end{aligned}$$

Substitute these back into our main equation (and assuming mixed partials are equal by Clairaut's Theorem, $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$):

$$\begin{aligned}\frac{\partial^2 \phi}{\partial \theta^2} &= -r \cos \theta \frac{\partial f}{\partial x} - r \sin \theta \left[-r \sin \theta \frac{\partial^2 f}{\partial x^2} + r \cos \theta \frac{\partial^2 f}{\partial x \partial y} \right] \\ &\quad - r \sin \theta \frac{\partial f}{\partial y} + r \cos \theta \left[-r \sin \theta \frac{\partial^2 f}{\partial x \partial y} + r \cos \theta \frac{\partial^2 f}{\partial y^2} \right]\end{aligned}$$

Expanding and grouping terms, we get the final formula:

$$\frac{\partial^2 \phi}{\partial \theta^2} = -r \cos \theta \frac{\partial f}{\partial x} - r \sin \theta \frac{\partial f}{\partial y} + r^2 \sin^2 \theta \frac{\partial^2 f}{\partial x^2} - 2r^2 \sin \theta \cos \theta \frac{\partial^2 f}{\partial x \partial y} + r^2 \cos^2 \theta \frac{\partial^2 f}{\partial y^2}$$

19. Proof: Continuous Partial Derivatives Imply Differentiability

Let $S \subseteq \mathbb{R}^n$, and $f : S \rightarrow \mathbb{R}$ be a scalar field. Let $a \in \text{Int}(S)$.

Theorem : Sufficient Condition for Differentiability

Assume that the partial derivatives $\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}$ exist at every point in some n -ball $B_r(a)$ (for some $r > 0$) and are all continuous at a . Then f is differentiable at a .

Proof:

We must show that there exists a linear map $T_a : \mathbb{R}^n \rightarrow \mathbb{R}$ such that:

$$f(a + v) - f(a) = T_a(v) + \|v\|E_a(v)$$

where $E_a(v) \rightarrow 0$ as $v \rightarrow 0$. If f is differentiable, we know $T_a(v) = \nabla f(a) \cdot v$. So we will prove that the following error term goes to 0:

$$E_a(v) = \begin{cases} \frac{f(a+v) - f(a) - \nabla f(a) \cdot v}{\|v\|} & \text{if } v \neq 0 \\ 0 & \text{if } v = 0 \end{cases}$$

Let $v \in B_r(0)$ such that $v \neq 0$. Let $\lambda = \|v\|$ and define the unit vector $u = v/\lambda$, so $v = \lambda u$ and $\|u\| = 1$. Let $u = u_1 e_1 + \dots + u_n e_n$.

We construct a sequence of vectors to build a "staircase" from a to $a + v$:

$$V_0 = 0, \quad V_1 = u_1 e_1, \quad V_2 = u_1 e_1 + u_2 e_2, \quad \dots \quad V_n = u$$

The total change in f can be written as a telescoping sum:

$$f(a + v) - f(a) = f(a + \lambda u) - f(a) = \sum_{i=1}^n [f(a + \lambda V_i) - f(a + \lambda V_{i-1})]$$

Let $b_i = a + \lambda V_{i-1}$. Note that $a + \lambda V_i = b_i + \lambda u_i e_i$. The sum becomes

$$\sum_{i=1}^n [f(b_i + \lambda u_i e_i) - f(b_i)].$$

We then apply the 1D Mean Value Theorem to each segment.

$$f(b_i + \lambda u_i e_i) - f(b_i) = \lambda u_i \frac{\partial f}{\partial x_i}(c_i)$$

where c_i is a point on the open line segment joining b_i and $b_i + \lambda u_i e_i$. Thus, $f(a + v) - f(a) = \lambda \sum_{i=1}^n \frac{\partial f}{\partial x_i}(c_i) u_i$.

Now substitute this into the numerator of $E_a(v)$, noting that $\lambda = \|v\|$:

$$\|v\|E_a(v) = \lambda \sum_{i=1}^n \frac{\partial f}{\partial x_i}(c_i) u_i - \lambda \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a) u_i = \|v\| \sum_{i=1}^n \left(\frac{\partial f}{\partial x_i}(c_i) - \frac{\partial f}{\partial x_i}(a) \right) u_i$$

Dividing by $\|v\|$:

$$E_a(v) = \sum_{i=1}^n \left(\frac{\partial f}{\partial x_i}(c_i) - \frac{\partial f}{\partial x_i}(a) \right) u_i$$

As $v \rightarrow 0$, we have $\lambda \rightarrow 0$. This forces $b_i \rightarrow a$ and therefore $c_i \rightarrow a$. Because the partial derivatives are **continuous** at a , $\lim_{c_i \rightarrow a} \frac{\partial f}{\partial x_i}(c_i) = \frac{\partial f}{\partial x_i}(a)$. Therefore, every term in the sum goes to 0, which implies $E_a(v) \rightarrow 0$. This completes the proof. $\blacksquare$

20. Equality of Mixed Partial Derivatives

In general, is $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$?

Counterexample where Mixed Partial Derivatives are NOT Equal

Consider $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by:

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Let $g(x, y) = \frac{\partial f}{\partial y}(x, y)$. We evaluate $g(x, 0)$ for $x \neq 0$:

$$g(x, 0) = \lim_{h \rightarrow 0} \frac{f(x, h) - f(x, 0)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{xh(x^2 - h^2)}{x^2 + h^2} \right) = \frac{x^3}{x^2} = x$$

Also $g(0, 0) = 0$. So $g(x, 0) = x$ for all $x \in \mathbb{R}$. Now we evaluate the mixed partial at the origin:

$$\frac{\partial^2 f}{\partial x \partial y}(0, 0) = \frac{\partial g}{\partial x}(0, 0) = \lim_{k \rightarrow 0} \frac{g(k, 0) - g(0, 0)}{k} = \lim_{k \rightarrow 0} \frac{k - 0}{k} = 1$$

However, by symmetry, evaluating the other mixed partial yields:

$$\frac{\partial^2 f}{\partial y \partial x}(0, 0) = -1$$

Since $1 \neq -1$, the order of differentiation matters for this function!

To guarantee equality, we need a continuity condition.

Theorem : Clairaut's Theorem (Schwarz's Theorem)

Let $f : S \rightarrow \mathbb{R}$ for $S \subseteq \mathbb{R}^2$. Let $(a, b) \in \text{Int}(S)$. Suppose the partial derivatives $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial^2 f}{\partial x \partial y}, \frac{\partial^2 f}{\partial y \partial x}$ exist at every point in some ball $B_r(a, b)$, and the mixed partials $\frac{\partial^2 f}{\partial x \partial y}$ and $\frac{\partial^2 f}{\partial y \partial x}$ are **continuous** at (a, b) . Then:

$$\frac{\partial^2 f}{\partial x \partial y}(a, b) = \frac{\partial^2 f}{\partial y \partial x}(a, b)$$

Proof:

Choose a rectangle $R(h, k)$ with vertices $(a, b), (a + h, b), (a, b + k), (a + h, b + k)$ such that it is strictly contained within $B_r(a, b)$. Define the double difference operator:

$$\Delta(h, k) = f(a + h, b + k) - f(a + h, b) - f(a, b + k) + f(a, b)$$

Let $G(x) = f(x, b + k) - f(x, b)$. Then $\Delta(h, k) = G(a + h) - G(a)$. Applying the Mean Value Theorem to G on $[a, a + h]$, there exists $x_0 \in (a, a + h)$ such that:

$$\Delta(h, k) = hG'(x_0) = h \left(\frac{\partial f}{\partial x}(x_0, b + k) - \frac{\partial f}{\partial x}(x_0, b) \right)$$

Now apply the MVT again to the function $\frac{\partial f}{\partial x}(x_0, y)$ with respect to y on $[b, b + k]$. There

exists $y_0 \in (b, b+k)$ such that:

$$\Delta(h, k) = hk \frac{\partial^2 f}{\partial y \partial x}(x_0, y_0)$$

where (x_0, y_0) is a point inside the rectangle $R(h, k)$.

By entirely symmetric logic, if we defined $H(y) = f(a+h, y) - f(a, y)$, we would find another point $(x_1, y_1) \in R(h, k)$ such that:

$$\Delta(h, k) = hk \frac{\partial^2 f}{\partial x \partial y}(x_1, y_1)$$

Equating the two expressions and dividing by hk :

$$\frac{\partial^2 f}{\partial y \partial x}(x_0, y_0) = \frac{\partial^2 f}{\partial x \partial y}(x_1, y_1)$$

Take the limit as $(h, k) \rightarrow (0, 0)$. Both (x_0, y_0) and (x_1, y_1) are trapped in the shrinking rectangle and must converge to (a, b) . Because the mixed partials are **continuous** at (a, b) , the limits evaluate to the functional values at (a, b) , completing the proof. ■

21. Extrema and The Hessian Matrix

Let $S \subseteq \mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$. For a point $a \in \text{Int}(S)$:

- $f(a) \geq f(x)$ for all $x \in B_r(a) \implies$ relative maximum.
- $f(a) \leq f(x)$ for all $x \in B_r(a) \implies$ relative minimum.
- If every neighborhood $B_r(a)$ contains points where $f(x) > f(a)$ AND points where $f(x) < f(a) \implies$ saddle point.

A necessary condition for a relative extremum (if f is differentiable) is $\nabla f(a) = 0$. Such points are called stationary points. To classify these points, we generalize the 1D second derivative test using the matrix of second partial derivatives.

Definition : The Hessian Matrix

If all second partial derivatives exist, the **Hessian Matrix** $H_f(a)$ is the $n \times n$ matrix defined by:

$$H_f(a) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(a) \right)_{1 \leq i, j \leq n} = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2}(a) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(a) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(a) & \frac{\partial^2 f}{\partial x_2^2}(a) \end{pmatrix} \quad (\text{for } n = 2)$$

If the mixed partials are continuous, by Clairaut's Theorem, $H_f(a)$ is a **real symmetric matrix**.

Definiteness and Classification

Because $H_f(a)$ is a real symmetric matrix, it has real eigenvalues λ_i , and an orthogonal basis of eigenvectors exists. We classify the critical point by the *definiteness* of $H_f(a)$:

- **Positive Definite** ($y^T H_f(a) y > 0$ for all $y \neq 0$): All eigenvalues are positive ($\lambda_i > 0$). The function curves upward in all directions. $\implies$ **Relative Minimum**.
- **Negative Definite**: All eigenvalues are negative ($\lambda_i < 0$). $\implies$ **Relative Maximum**.
- **Indefinite**: Eigenvalues have mixed signs (some positive, some negative). $\implies$ **Saddle Point**.

22. Extrema, Stationary Points, and Saddle Points

Let $S \subseteq \mathbb{R}^n$, $f : S \rightarrow \mathbb{R}$, and $a \in S$. We classify the behavior of f at a as follows:

Definition : Extrema and Saddle Points

1. f has an **absolute maximum** at a if $f(x) \leq f(a)$ for all $x \in S$.
2. f has an **absolute minimum** at a if $f(x) \geq f(a)$ for all $x \in S$.
3. f has a **relative maximum** at a if there exists an n -ball $B_r(a) \subseteq S$ such that $f(x) \leq f(a)$ for all $x \in B_r(a)$.
4. f has a **relative minimum** at a if there exists an n -ball $B_r(a) \subseteq S$ such that $f(x) \geq f(a)$ for all $x \in B_r(a)$.

If f has a relative maximum or relative minimum at $a \in S$, then f is said to have an **extremum** at a . Note that an extremum implies $a \in \text{Int}(S)$.

Note: Suppose $a \in \text{Int}(S)$, f has an extremum at a , and f is differentiable at a . Then $\frac{\partial f}{\partial x_i}(a) = 0$ for all $1 \leq i \leq n$. That is, $\nabla f(a) = (0, \dots, 0)$.

Definition : Stationary Point

If f is differentiable at $a \in \text{Int}(S)$ and $\nabla f(a) = 0$, then a is called a **stationary point** of f . (Thus, if f has an extremum at a and is differentiable, it *MUST* be a stationary point).

Definition : Saddle Point

Let $a \in \text{Int}(S)$ be a stationary point of f (so $\nabla f(a) = 0$). Then a is called a **saddle point** of f if every n -ball $B_r(a)$ contains points x, y such that:

$$f(x) < f(a) < f(y)$$

Thus, any stationary point of f is either a saddle point or a point of extremum (relative max/min).

Classic Examples of Stationary Points

1. Relative/Absolute Maximum: $f(x, y) = 2 - x^2 - y^2$.

$f(x, y) = 2 - (x^2 + y^2) \leq 2 = f(0, 0)$ for all $(x, y) \in \mathbb{R}^2$. Thus $(0, 0)$ is a relative and absolute maximum.

2. Relative/Absolute Minimum: $f(x, y) = x^2 + y^2$.

$f(x, y) \geq 0 = f(0, 0)$ for all $(x, y) \in \mathbb{R}^2$. Thus $(0, 0)$ is a relative and absolute minimum.

3. Standard Saddle Point: $f(x, y) = xy$.

$\nabla f(0, 0) = (y, x)|_{(0,0)} = (0, 0)$. In any $B_r(0)$, there exist points where $xy > 0$ (Q1, Q3) and points where $xy < 0$ (Q2, Q4). Thus $(0, 0)$ is a saddle point.

4. The Monkey Saddle: $f(x, y) = x^3 - 3xy^2$.

$\nabla f(0, 0) = (3x^2 - 3y^2, -6xy)|_{(0,0)} = (0, 0)$. We can factor $f(x, y) = x(x^2 - 3y^2) = x(x - \sqrt{3}y)(x + \sqrt{3}y)$. In any $B_r(0)$, the sign of $f(x, y)$ alternates six times depending on the sector, guaranteeing points both strictly greater than and strictly less than $f(0, 0) = 0$. Thus $(0, 0)$ is a saddle point.

5. Non-Strict Extrema:

- $f(x, y) = x^2y^2 \geq 0 = f(0, 0)$. $(0, 0)$ is an absolute/relative minimum (though a flat valley along the axes).
- $f(x, y) = 1 - x^2 = 1 - x^2 \leq 1 = f(0, 0)$. $(0, 0)$ is an absolute/relative maximum (a ridge along the y-axis).

23. The Hessian Matrix and Second-Order Taylor Formula

Suppose f has continuous second-order partial derivatives on some $B_r(a)$, meaning $\frac{\partial^2 f}{\partial x_i \partial x_j}$ exists and is continuous at all points in $B_r(a)$.

Definition : The Hessian Matrix

We define the **Hessian matrix** of f at a to be the $n \times n$ real matrix:

$$H(a) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(a) \right)_{1 \leq i, j \leq n}$$

By Clairaut's Theorem (since the mixed partials are continuous), $H(a)$ is a **real symmetric** $n \times n$ **matrix**.

Theorem : Second-Order Taylor Formula for Scalar Fields

Suppose f has continuous second-order partial derivatives in an n -ball $B_r(a)$. Then for all $y \in B_r(0)$, we have:

$$f(a + y) - f(a) = \nabla f(a) \cdot y + \frac{1}{2} y^T H(a + cy) y$$

for some $c \in (0, 1)$.

Alternatively, we can write this as:

$$f(a + y) - f(a) = \nabla f(a) \cdot y + \frac{1}{2} y^T H(a) y + \|y\|^2 E_2(a, y)$$

where $E_2(a, y) \rightarrow 0$ as $y \rightarrow 0$.

Proof:

We will use the following theorem from one-variable calculus: *Suppose g is a real-valued function on an open interval J containing $[0, 1]$, and g is twice continuously differentiable. Then $g(1) - g(0) = g'(0) + \frac{1}{2}g''(c)$ for some $c \in (0, 1)$.*

Fix some $y \in B_r(0)$ and define $g(u) = f(a + uy)$ for $u \in [-1, 1]$. Then $g(1) = f(a + y)$ and $g(0) = f(a)$. Note that $g = f \circ r$, where $r(u) = a + uy$. By the Chain Rule, $g'(u) = \nabla f(r(u)) \cdot r'(u)$. Since $r'(u) = y$, we have:

$$g'(u) = \nabla f(a + uy) \cdot y = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a + uy) y_i$$

Thus, evaluated at $u = 0$: $g'(0) = \nabla f(a) \cdot y$.

Using the Chain Rule again to find the second derivative $g''(u)$:

$$\begin{aligned} g''(u) &= \frac{d}{du} \left(\sum_{i=1}^n \frac{\partial f}{\partial x_i}(r(u)) y_i \right) = \sum_{i=1}^n y_i \frac{d}{du} \left(\frac{\partial f}{\partial x_i}(r(u)) \right) \\ &= \sum_{i=1}^n y_i \left(\sum_{j=1}^n \frac{\partial^2 f}{\partial x_j \partial x_i}(r(u)) y_j \right) = \sum_{i=1}^n \sum_{j=1}^n y_i y_j \frac{\partial^2 f}{\partial x_i \partial x_j}(a + uy) \\ &= y^T H(a + uy) y \end{aligned}$$

Thus, $g''(c) = y^T H(a + cy) y$ for $0 < c < 1$. Substituting $g(1)$, $g(0)$, $g'(0)$, and $g''(c)$ into the 1D theorem yields the exact form:

$$f(a + y) - f(a) = \nabla f(a) \cdot y + \frac{1}{2} y^T H(a + cy) y$$

To isolate the error term, we define $E_2(a, y)$ such that:

$$E_2(a, y) = \begin{cases} \frac{1}{2} \frac{y^T [H(a+cy) - H(a)] y}{\|y\|^2} & \text{if } y \neq 0 \\ 0 & \text{if } y = 0 \end{cases}$$

We must show $E_2(a, y) \rightarrow 0$ as $y \rightarrow 0$. We use the fact that for any $n \times n$ matrix $A = (a_{ij})$, $|y^T A y| \leq \|y\|^2 (\sum \sum |a_{ij}|)$. Applying this to our error term:

$$\begin{aligned} |E_2(a, y)| &= \frac{1}{2 \|y\|^2} |y^T (H(a + cy) - H(a)) y| \\ &\leq \frac{1}{2 \|y\|^2} \|y\|^2 \sum_{i=1}^n \sum_{j=1}^n \left| \frac{\partial^2 f}{\partial x_i \partial x_j}(a + cy) - \frac{\partial^2 f}{\partial x_i \partial x_j}(a) \right| \\ &= \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \left| \frac{\partial^2 f}{\partial x_i \partial x_j}(a + cy) - \frac{\partial^2 f}{\partial x_i \partial x_j}(a) \right| \end{aligned}$$

As $y \rightarrow 0$, $cy \rightarrow 0$. Because the second partial derivatives are assumed to be **continuous** at a , the limit of the difference in the absolute value goes to zero. Hence $E_2(a, y) \rightarrow 0$, completing the proof. ■

24. The Second Derivative Test via Linear Algebra

Let $f : S \rightarrow \mathbb{R}$ have continuous second-order partial derivatives on $B_r(a) \subseteq S$. Let a be a **stationary point** of f (so $\nabla f(a) = 0$).

Substituting $\nabla f(a) = 0$ into our Taylor formula, the linear term vanishes, leaving:

$$f(a + y) - f(a) = \frac{1}{2}y^T H(a)y + \|y\|^2 E_2(a, y)$$

The behavior of the function near a is dominated by the quadratic form $y^T H(a)y$. Because $H(a)$ is a real symmetric matrix, we invoke a fundamental theorem from Linear Algebra:

Theorem : Spectral Theorem and Definiteness

Let A be a real $n \times n$ symmetric matrix.

1. A is **positive definite** ($y^T A y > 0$ for all $y \neq 0$) $\iff$ all eigenvalues of A are strictly positive.
2. A is **negative definite** ($y^T A y < 0$ for all $y \neq 0$) $\iff$ all eigenvalues of A are strictly negative.

This theorem depends on the fact that a real symmetric matrix has all real eigenvalues and an orthonormal basis of eigenvectors, allowing A to be diagonalized into a matrix D containing only the eigenvalues on the diagonal.

Conclusion for Extrema: If $H(a)$ is positive definite, then $f(a + y) - f(a) > 0$ for small y , meaning $f(a + y) > f(a)$ (Relative Minimum). If $H(a)$ is negative definite, $f(a + y) < f(a)$ (Relative Maximum). If the eigenvalues have mixed signs, a is a Saddle Point.

25. Proof of the Second Derivative Test

Theorem : Eigenvalues and Extrema

Let $S \subseteq \mathbb{R}^n$, $f : S \rightarrow \mathbb{R}$, $a \in \text{Int}(S)$. Suppose f has continuous second-order partial derivatives on some $B_r(a) \subseteq S$. Assume a is a stationary point of f ($\nabla f(a) = 0$).

- (a) If all eigenvalues of $H(a)$ are positive, f has a relative minimum at a .
- (b) If all eigenvalues of $H(a)$ are negative, f has a relative maximum at a .
- (c) If $H(a)$ has both positive and negative eigenvalues, f has a saddle point at a .

Proof:

Since a is a stationary point, the First-order term vanishes, and the Taylor formula gives:

$$f(a + y) - f(a) = \frac{1}{2}y^T H(a)y + \|y\|^2 E_2(a, y)$$

where $E_2(a, y) \rightarrow 0$ as $y \rightarrow 0$ (for all $y \in B_r(0)$). We claim that there exists $\epsilon > 0$ such that if $0 < \|y\| < \epsilon$, then $f(a + y) - f(a)$ has the exact same sign as the quadratic form $y^T H(a)y$.

Part (a): Relative Minimum

Assume all eigenvalues of $H(a)$ are positive. Let these eigenvalues be $\lambda_1, \dots, \lambda_n$. Let $h = \min(\lambda_i)$. Since all $\lambda_i > 0$, we have $h > 0$. For any $0 \leq u < h$, we have $\lambda_i - u > 0$ for all i . Hence, the real symmetric matrix $(H(a) - uI)$ is positive definite. This means:

$$y^T(H(a) - uI)y > 0 \quad \forall y \in \mathbb{R}^n \setminus \{0\}$$

Distributing the y , we get $y^T H(a)y > y^T(uI)y = u\|y\|^2$. Choose $u = h/2$. Then:

$$y^T H(a)y > \frac{h}{2}\|y\|^2 \quad \forall y \neq 0$$

Now, because $E_2(a, y) \rightarrow 0$ as $y \rightarrow 0$, there exists an $\epsilon > 0$ (where $\epsilon < r$) such that $|E_2(a, y)| < \frac{h}{4}$ for all $0 < \|y\| < \epsilon$. Thus, $\|y\|^2 |E_2(a, y)| < \frac{h}{4}\|y\|^2$.

Substitute these bounds back into the Taylor expansion:

$$\begin{aligned} f(a+y) - f(a) &= \frac{1}{2}y^T H(a)y + \|y\|^2 E_2(a, y) \\ &\geq \frac{1}{2}y^T H(a)y - \|y\|^2 |E_2(a, y)| \\ &> \frac{1}{2} \left(\frac{h}{2}\|y\|^2 \right) - \frac{h}{4}\|y\|^2 \\ &= \frac{h}{4}\|y\|^2 - \frac{h}{4}\|y\|^2 = 0 \end{aligned}$$

Since $f(a+y) - f(a) > 0$ for all $y \in B_\epsilon(0) \setminus \{0\}$, f has a **relative minimum** at a .

Part (b): Relative Maximum

Apply Part (a) to the function $-f$. Since $H_{-f}(a) = -H_f(a)$, if all eigenvalues of $H_f(a)$ are negative, all eigenvalues of $H_{-f}(a)$ are positive. By Part (a), $-f$ has a relative minimum at a , which means $(-f)(a+y) > (-f)(a)$. Multiplying by -1 flips the inequality: $f(a+y) < f(a)$. Thus, f has a **relative maximum**.

Part (c): Saddle Point

Suppose λ_1 and λ_2 are two eigenvalues of $H(a)$ of opposite signs. Let $h = \min(|\lambda_1|, |\lambda_2|)$. For $u \in (-h, h)$, the shifted eigenvalues $\lambda_1 - u$ and $\lambda_2 - u$ remain of opposite signs. Thus, the quadratic form $y^T(H(a) - uI)y$ takes both positive and negative values in every neighborhood of 0.

There exists some direction y where $y^T(H(a) - uI)y > 0 \implies y^T H(a)y > u\|y\|^2$. Choosing $u = h/2$, we get $y^T H(a)y > \frac{h}{2}\|y\|^2$. Using the exact same bounding logic as in Part (a), $f(a+y) - f(a) > 0$ along this path.

Conversely, there exists another direction y where $y^T(H(a) - uI)y < 0 \implies y^T H(a)y < u\|y\|^2$. Choosing $u = -h/2$, we get $y^T H(a)y < -\frac{h}{2}\|y\|^2$. Bounding the error term gives:

$$f(a+y) - f(a) \leq \frac{1}{2}y^T H(a)y + \|y\|^2 |E_2(a, y)| < -\frac{h}{4}\|y\|^2 + \frac{h}{4}\|y\|^2 = 0$$

Since f takes both strictly greater and strictly lesser values in every neighborhood of a , it is a **saddle point**. ■

26. The 2D Second Derivative Test ($AC - B^2$)

For functions of two variables ($S \subseteq \mathbb{R}^2$), we can determine the definiteness of the Hessian matrix directly using its determinant, avoiding the explicit computation of eigenvalues.

Theorem : The 2-Variable Extrema Test

Let $f : S \rightarrow \mathbb{R}$ ($S \subseteq \mathbb{R}^2$) have continuous second-order partial derivatives. Let $a \in \text{Int}(S)$ be a stationary point of f ($\nabla f(a) = 0$).

Let the entries of the Hessian matrix $H(a)$ be denoted as:

$$A = \frac{\partial^2 f}{\partial x^2}(a), \quad B = \frac{\partial^2 f}{\partial x \partial y}(a), \quad C = \frac{\partial^2 f}{\partial y^2}(a)$$

Define the discriminant Δ as the determinant of the Hessian:

$$\Delta = \det(H(a)) = \det \begin{pmatrix} A & B \\ B & C \end{pmatrix} = AC - B^2$$

Then we can classify the stationary point as follows:

1. If $\Delta < 0 \implies f$ has a **saddle point** at a .
2. If $\Delta > 0$ and $A > 0 \implies f$ has a **relative minimum** at a .
3. If $\Delta > 0$ and $A < 0 \implies f$ has a **relative maximum** at a .
4. If $\Delta = 0 \implies$ the test is **inconclusive** (higher-order Taylor terms are needed).

27. Taylor's Theorem in One Variable (Ross)

Let $f : (a, b) \rightarrow \mathbb{R}$ be a real-valued function of one variable, and let $c \in (a, b)$. Suppose f has derivatives of all orders at c , that is, $f^{(n)}(c)$ exists for all $n \in \mathbb{N}$.

Definition : Taylor Series and Remainder

For any $x \in (a, b)$, we consider the series:

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(c)}{k!} (x - c)^k$$

This is called the **Taylor series** for f about c . For a particular x , this series may or may not converge.

For $n \geq 1$, we define the **remainder** $R_n(x)$ by:

$$R_n(x) = f(x) - \sum_{k=0}^{n-1} \frac{f^{(k)}(c)}{k!} (x - c)^k$$

Note that $\lim_{n \rightarrow \infty} R_n(x) = 0 \iff f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(c)}{k!} (x - c)^k$.

Theorem : Taylor's Theorem

Let $f : (a, b) \rightarrow \mathbb{R}$ and $c \in (a, b)$ (allowing $a = -\infty, b = \infty$). Suppose the n -th derivative $f^{(n)}$ exists on (a, b) . Then for each $x \neq c$ in (a, b) , there exists a point y strictly between c and x such that:

$$R_n(x) = \frac{f^{(n)}(y)}{n!}(x - c)^n$$

Thus, $f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(c)}{k!}(x - c)^k + \frac{f^{(n)}(y)}{n!}(x - c)^n$.

Proof:

Fix $x \neq c$ and fix $n \geq 1$. Let M be the unique real number which satisfies:

$$f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(c)}{k!}(x - c)^k + \frac{M}{n!}(x - c)^n$$

We need to show $M = f^{(n)}(y)$ for some y between c and x .

Define a function $g : (a, b) \rightarrow \mathbb{R}$ by:

$$g(t) = \sum_{k=0}^{n-1} \frac{f^{(k)}(c)}{k!}(t - c)^k + \frac{M}{n!}(t - c)^n - f(t)$$

By construction, evaluating at $t = c$ gives $g(c) = 0$, and taking derivatives gives $g^{(k)}(c) = 0$ for all $1 \leq k < n$. Furthermore, our choice of M guarantees that $g(x) = 0$.

Because $g(c) = g(x) = 0$, **Rolle's Theorem** implies there exists x_1 between c and x such that $g'(x_1) = 0$. Now we have $g'(c) = 0$ and $g'(x_1) = 0$. Applying Rolle's Theorem to g' , there exists x_2 between c and x_1 such that $g''(x_2) = 0$.

We proceed inductively. Because $g^{(n-1)}(c) = 0$ and $g^{(n-1)}(x_{n-1}) = 0$, there exists x_n between c and x_{n-1} such that $g^{(n)}(x_n) = 0$.

Let $y = x_n$. Then y lies strictly between c and x , and $g^{(n)}(y) = 0$. Now, let us explicitly calculate the n -th derivative of our original function $g(t)$. The polynomial terms of degree less than n vanish, leaving:

$$g^{(n)}(t) = M - f^{(n)}(t)$$

Evaluating at $t = y$:

$$0 = g^{(n)}(y) = M - f^{(n)}(y) \implies M = f^{(n)}(y)$$

This completes the proof. ■

Theorem : Convergence Corollary

If $f^{(n)}$ exists on (a, b) for all $n \in \mathbb{N}$ and the derivatives are uniformly bounded by a constant C (i.e., $|f^{(n)}(x)| \leq C$ for all $x \in (a, b)$ and all $n \in \mathbb{N}$), then $\lim_{n \rightarrow \infty} R_n(x) = 0$. Therefore, the Taylor series converges to $f(x)$.

Proof:

For any $x \in (a, b)$, by Taylor's Theorem:

$$|R_n(x)| = \left| \frac{f^{(n)}(y)}{n!} (x - c)^n \right| \leq \frac{C}{n!} |x - c|^n \quad \text{for all } n$$

It is a standard limit that for any fixed z , $\frac{z^n}{n!} \rightarrow 0$ as $n \rightarrow \infty$. Thus, $R_n(x) \rightarrow 0$. ■

28. Classic Applications of Taylor Series

Example 1: The Exponential Function

Let $f(x) = e^x$ for $x \in \mathbb{R}$. Then $f^{(n)}(x) = e^x$ for all $n \in \mathbb{N}$. Evaluated at $c = 0$, $f^{(n)}(0) = 1$. The Taylor series is $\sum_{k=0}^{\infty} \frac{x^k}{k!}$.

Consider any bounded interval $(-M, M) \subseteq \mathbb{R}$. All derivatives of e^x are bounded by the constant e^M on this interval. By the Convergence Corollary, the remainder goes to 0. Hence, $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$ for all $x \in \mathbb{R}$.

Example 2: The Sine Function

Let $f(x) = \sin x$. The derivatives cycle: $\cos x, -\sin x, -\cos x, \sin x$. Evaluated at $c = 0$, the pattern is $1, 0, -1, 0$. The Taylor series is:

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1}$$

Since $|\sin x| \leq 1$ and $|\cos x| \leq 1$, the derivatives are uniformly bounded by $C = 1$ on all of $\mathbb{R}$. Thus, the series converges to $\sin x$ for all $x \in \mathbb{R}$.

Example 3: Proving the Alternating Harmonic Series

We will prove that $\log_e(2) = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$.

Let $f(x) = \log_e(1+x)$ for $x \in (-1, \infty)$. The derivatives are:

$$f'(x) = \frac{1}{1+x}, \quad f''(x) = \frac{-1}{(1+x)^2}, \quad f^{(3)}(x) = \frac{2}{(1+x)^3}$$

In general, $f^{(n)}(x) = \frac{(-1)^{n+1}(n-1)!}{(1+x)^n}$. Evaluated at $c = 0$, $f^{(n)}(0) = (-1)^{n+1}(n-1)!$. The Taylor series for f about 0 is:

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}(k-1)!}{k!} x^k = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} x^k = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

Now, apply Taylor's Theorem with interval $(-1, \infty)$, center $c = 0$, and evaluate at $x = 1$: For all n , there exists $y_n \in (0, 1)$ such that:

$$R_n(1) = \frac{f^{(n)}(y_n)}{n!} (1-0)^n = \frac{(-1)^{n+1}(n-1)!}{n!(1+y_n)^n} = \frac{(-1)^{n+1}}{n(1+y_n)^n}$$

Taking the absolute value:

$$|R_n(1)| = \frac{1}{n(1+y_n)^n}$$

Because $y_n > 0$, the term $(1+y_n)^n > 1$. Therefore:

$$|R_n(1)| < \frac{1}{n} \quad \text{for all } n$$

As $n \rightarrow \infty$, $\frac{1}{n} \rightarrow 0$. By the Squeeze Theorem, $\lim_{n \rightarrow \infty} R_n(1) = 0$. Because the remainder vanishes at $x = 1$, the series converges exactly to $f(1) = \log_e(2)$.

29. Review: The Second-Order Taylor Formula

Recall: First-Order Taylor Formula

Recall that a scalar field $f : S \rightarrow \mathbb{R}$ (where $S \subseteq \mathbb{R}^n$) is differentiable at $a \in \text{Int}(S)$ if there exists a linear map $T_a : \mathbb{R}^n \rightarrow \mathbb{R}$ such that for all $v \in B_r(0)$, we have the first-order Taylor formula:

$$f(a+v) = f(a) + T_a(v) + \|v\|E_a(v)$$

where $E_a(v) \rightarrow 0$ as $v \rightarrow 0$.

The 1D Analogue ($n = 1$): For $S \subseteq \mathbb{R}$, $f(a+v) = f(a) + f'(a)v + |v|E_a(v)$. From single-variable calculus, the exact second-order expansion is:

$$f(a+v) = f(a) + f'(a)v + \frac{f''(b)}{2}v^2$$

Letting $x = a+v \implies v = x-a$, we get the familiar form:

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(b)}{2}(x-a)^2$$

The Hessian Matrix

Suppose f has continuous second-order partial derivatives $\frac{\partial^2 f}{\partial x_i \partial x_j}$ on some n -ball $B_r(a)$ (for $r > 0$) for all $1 \leq i, j \leq n$. From Clairaut's Theorem (proved previously), we know $\frac{\partial^2 f}{\partial x_i \partial x_j}(a) = \frac{\partial^2 f}{\partial x_j \partial x_i}(a)$.

We can form the $n \times n$ real matrix called the **Hessian Matrix**, $H_f(a)$:

$$H_f(a) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(a) \right)_{1 \leq i, j \leq n}$$

The 3×3 Hessian Matrix

For $n = 3$, the Hessian is a real symmetric 3×3 matrix:

$$H_f(a) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_1 \partial x_3} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \frac{\partial^2 f}{\partial x_2 \partial x_3} \\ \frac{\partial^2 f}{\partial x_3 \partial x_1} & \frac{\partial^2 f}{\partial x_3 \partial x_2} & \frac{\partial^2 f}{\partial x_3^2} \end{pmatrix}$$

Theorem : Second-Order Taylor Formula

Let $S \subseteq \mathbb{R}^n$, $f : S \rightarrow \mathbb{R}$ be a scalar field, and $a \in \text{Int}(S)$. Suppose f has continuous second-order partial derivatives at every point of some $B_r(a)$ for $r > 0$. Then for all $y \in B_r(0)$ (so $a + y \in B_r(a)$), we have:

$$f(a + y) = f(a) + \nabla f(a) \cdot y + \frac{1}{2} y^T H_f(a) y + \|y\|^2 E_2(a, y)$$

where $E_2(a, y) \rightarrow 0$ as $y \rightarrow 0$.

Note: The quadratic form $y^T H_f(a) y$ explicitly expands to $\sum_{j=1}^n \sum_{i=1}^n y_i y_j \frac{\partial^2 f}{\partial x_i \partial x_j}(a)$.

Proof:

We will use Taylor's Theorem in one variable. Let $g : (a, b) \rightarrow \mathbb{R}$ be continuously differentiable twice on (a, b) . If $[0, 1] \subseteq (a, b)$, then:

$$g(1) - g(0) = g'(0) + \frac{g''(c)}{2} \quad \text{for some } c \in (0, 1)$$

Let $y \in B_r(0)$. Define the 1D "slice" function $g(u) = f(a + uy)$ for $u \in [-1, 1]$. Note that $g(1) = f(a + y)$ and $g(0) = f(a)$.

We write g as a composition: $g(u) = f(r(u))$ where $r(u) = a + uy$. Here, $r : \mathbb{R} \rightarrow \mathbb{R}^n$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$. By the Chain Rule, $g'(u) = \nabla f(r(u)) \cdot r'(u)$. Since $r(u) = (a_1 + uy_1, \dots, a_n + uy_n)$, we have $r'(u) = (y_1, \dots, y_n) = y$. Thus:

$$g'(u) = \nabla f(a + uy) \cdot y = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a + uy) y_i$$

Evaluating at $u = 0$ yields the first derivative term: $g'(0) = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a) y_i = \nabla f(a) \cdot y$.

To find the second derivative $g''(u)$, we differentiate $g'(u)$ with respect to u :

$$\begin{aligned} g''(u) &= \frac{d}{du} \left(\sum_{i=1}^n \frac{\partial f}{\partial x_i}(a + uy) y_i \right) = \sum_{i=1}^n y_i \frac{d}{du} \left(\frac{\partial f}{\partial x_i}(a + uy) \right) \\ &= \sum_{i=1}^n y_i \left(\sum_{j=1}^n \frac{\partial^2 f}{\partial x_j \partial x_i}(a + uy) y_j \right) \\ &= \sum_{i=1}^n \sum_{j=1}^n \frac{\partial^2 f}{\partial x_j \partial x_i}(a + uy) y_i y_j = y^T H_f(a + uy) y \end{aligned}$$

Now, substitute $g(1), g(0), g'(0)$, and $g''(c)$ back into the 1D Taylor formula:

$$f(a + y) - f(a) = \nabla f(a) \cdot y + \frac{1}{2} y^T H_f(a + cy) y$$

To isolate the Hessian at a , we add and subtract $\frac{1}{2} y^T H_f(a) y$:

$$f(a + y) = f(a) + \nabla f(a) \cdot y + \frac{1}{2} y^T H_f(a) y + \frac{1}{2} y^T (H_f(a + cy) - H_f(a)) y$$

We define the remainder term as $\|y\|^2 E_2(a, y) = \frac{1}{2} y^T (H_f(a + cy) - H_f(a)) y$. Because the second partial derivatives are continuous, the matrix $(H_f(a + cy) - H_f(a))$ approaches the zero matrix as $y \rightarrow 0$, forcing $E_2(a, y) \rightarrow 0$. ■

30. Rigorous Foundation of the Second Derivative Test

Bounding the Quadratic Error Term

Recall our Second-Order Taylor formula for a scalar field f at $a \in \text{Int}(S)$:

$$f(a + y) = f(a) + \nabla f(a) \cdot y + \frac{1}{2} y^T H_f(a) y + \|y\|^2 E_2(a, y)$$

where the remainder term was defined as:

$$\|y\|^2 E_2(a, y) = \frac{1}{2} y^T (H_f(a + cy) - H_f(a)) y \quad \text{for some } c \in (0, 1)$$

Let $A = H_f(a + cy) - H_f(a)$ be an $n \times n$ real symmetric matrix. We want to show that $|y^T A y|$ is bounded. Using the Cauchy-Schwarz inequality:

$$|y^T A y| = |y \cdot (A y)| \leq \|y\| \|A y\|$$

Let $A = (a_{ij})$. The i -th component of the vector $A y$ is $\sum_{j=1}^n a_{ij} y_j$. Thus:

$$\|A y\| \leq \|y\| \sum_{i=1}^n \sum_{j=1}^n |a_{ij}|$$

Combining these gives the crucial bound:

$$|y^T A y| \leq \|y\|^2 \sum_{i=1}^n \sum_{j=1}^n |a_{ij}|$$

Applying this to our error term:

$$|E_2(a, y)| \leq \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \left| \frac{\partial^2 f}{\partial x_i \partial x_j}(a + cy) - \frac{\partial^2 f}{\partial x_i \partial x_j}(a) \right|$$

As $y \rightarrow 0$, $a + cy \rightarrow a$. Since the second partial derivatives are continuous, the difference goes to 0. Thus, $E_2(a, y) \rightarrow 0$.

31. The Spectral Theorem and Definiteness

To understand the behavior of the quadratic form $\frac{1}{2}y^T H_f(a)y$, we must rely on a fundamental theorem from Linear Algebra.

Theorem : Orthogonal Diagonalization (Spectral Theorem)

Let $A = (a_{ij})$ be a real symmetric $n \times n$ matrix. Then there exists an orthonormal basis of $\mathbb{R}^n$ consisting entirely of eigenvectors of A .

Equivalently, there exists an orthogonal matrix S (where $S^{-1} = S^T$) such that:

$$S^T A S = D = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

where $\lambda_1, \dots, \lambda_n$ are the real eigenvalues of A .

Using this theorem, we can easily prove the conditions for positive and negative definiteness.

Theorem : Definiteness via Eigenvalues

Let A be a real symmetric $n \times n$ matrix.

- (a) A is positive definite ($y^T A y > 0$ for all $y \neq 0$) $\iff$ all eigenvalues of A are positive ($\lambda_i > 0$).
- (b) A is negative definite ($y^T A y < 0$ for all $y \neq 0$) $\iff$ all eigenvalues of A are negative ($\lambda_i < 0$).

Proof:

We know $\exists$ an orthogonal matrix S such that $S^T A S = D$. Since S is orthogonal, $S S^T = I$, meaning we can insert it into our quadratic form:

$$y^T A y = y^T (S D S^T) y = (S^T y)^T D (S^T y)$$

Let $x = S^T y$. Because S is invertible, $x = 0 \iff y = 0$. The quadratic form simplifies to:

$$y^T A y = x^T D x = (x_1, \dots, x_n) \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \lambda_1 x_1^2 + \dots + \lambda_n x_n^2$$

($\Leftarrow$) If all $\lambda_i > 0$, then for any $x \neq 0$, the sum $\sum \lambda_i x_i^2 > 0$. Thus $y^T A y > 0$.

($\Rightarrow$) Suppose A is positive definite, so $y^T A y > 0$ for all $y \neq 0$. Assume for contradiction

that some $\lambda_k \leq 0$. Choose the vector x such that $x_k = 1$ and all other $x_i = 0$. Then $y^T Ay = \lambda_k(1)^2 = \lambda_k \leq 0$, which contradicts positive definiteness. Thus, all $\lambda_i > 0$. The proof for negative definiteness is exactly analogous. ■

32. Classification of Stationary Points

Definition : Extrema and Stationary Points

Let $S \subseteq \mathbb{R}^n$, $f : S \rightarrow \mathbb{R}$, and $a \in S$.

- f has an **absolute maximum (minimum)** at a if $f(x) \leq f(a)$ (or $f(x) \geq f(a)$) for all $x \in S$.
- f has a **relative maximum (minimum)** at a if there exists $B_r(a) \subseteq S$ such that $f(x) \leq f(a)$ (or $f(x) \geq f(a)$) for all $x \in B_r(a)$.

If $a \in \text{Int}(S)$ and f is differentiable at a , then having an extremum implies $\nabla f(a) = 0$. A point where $\nabla f(a) = 0$ is called a **stationary point**.

If a is a stationary point, but in every neighborhood $B_r(a)$ there exist points where $f(x) < f(a)$ AND points where $f(x) > f(a)$, then a is called a **saddle point**.

Theorem : Classification via Hessian Eigenvalues

Suppose f has continuous second-order partial derivatives in $B_r(a)$. Suppose a is a stationary point ($\nabla f(a) = 0$). We look at the eigenvalues of the Hessian $H_f(a)$:

1. If all eigenvalues are positive, f has a **relative minimum** at a .
2. If all eigenvalues are negative, f has a **relative maximum** at a .
3. If there is at least one positive eigenvalue and at least one negative eigenvalue, f has a **saddle point** at a .

(If all eigenvalues are non-zero but some are zero, the test is inconclusive).

Corollary: The 2D Test ($AC - B^2$)

Let $S \subseteq \mathbb{R}^2$. Let $\nabla f(a) = 0$. The Hessian is:

$$H_f(a) = \begin{pmatrix} A & B \\ B & C \end{pmatrix} \quad \text{where } A = \frac{\partial^2 f}{\partial x^2}, B = \frac{\partial^2 f}{\partial x \partial y}, C = \frac{\partial^2 f}{\partial y^2}$$

Let $\Delta = \det(H_f(a)) = AC - B^2$. Since the determinant is the product of the eigenvalues ($\Delta = \lambda_1 \lambda_2$):

- $\lambda_1 > 0, \lambda_2 > 0 \iff \Delta > 0 \text{ and } A > 0 \implies \text{Relative Minimum}$
- $\lambda_1 < 0, \lambda_2 < 0 \iff \Delta > 0 \text{ and } A < 0 \implies \text{Relative Maximum}$
- $\lambda_1, \lambda_2 \text{ have opposite signs} \iff \Delta < 0 \implies \text{Saddle Point}$
- $\Delta = 0 \implies \text{Test is Inconclusive}$

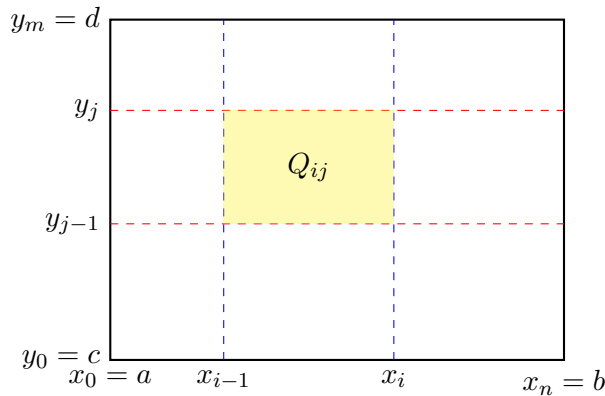
33. Multiple Integrals: Partitions and Step Functions

Reference: Apostol Calculus Volume 2, Chapter 11

We seek to rigorously define the double integral of a scalar field $f : Q \rightarrow \mathbb{R}$ over a closed rectangle $Q = [a, b] \times [c, d] \subseteq \mathbb{R}^2$.

Definition : Partitions and Subrectangles

Let $P_1 = \{x_0, x_1, \dots, x_n\}$ with $a = x_0 < x_1 < \dots < x_n = b$ be a partition of $[a, b]$.
 Let $P_2 = \{y_0, y_1, \dots, y_m\}$ with $c = y_0 < y_1 < \dots < y_m = d$ be a partition of $[c, d]$.
 The Cartesian product $P = P_1 \times P_2 = \{(x_i, y_j) \mid 0 \leq i \leq n, 0 \leq j \leq m\}$ is a **partition of Q** . It decomposes Q into mn subrectangles of the form $[x_{i-1}, x_i] \times [y_{j-1}, y_j]$.
 A partition P' of Q is said to be **finer** than P if every point of P is a point of P' ($P \subseteq P'$).



Definition : Step Functions on Rectangles

An **open subrectangle** of Q associated to P is of the form $(x_{i-1}, x_i) \times (y_{j-1}, y_j)$. It is

the subset with all its edges missing.

A function $f : Q \rightarrow \mathbb{R}$ is called a **step function** if there exists a partition $P = P_1 \times P_2$ of Q such that f is strictly **constant** on each of the open subrectangles associated to P . (The values f takes on the actual boundary edges do not matter in integration theory).

34. The Double Integral of a Step Function

Definition : Double Integral of a Step Function

Let $f : Q \rightarrow \mathbb{R}$ be a step function with respect to P . Suppose f takes the constant value c_{ij} on the open subrectangle $Q_{ij} = (x_{i-1}, x_i) \times (y_{j-1}, y_j)$. Then we define the **double integral** of f over Q to be:

$$\iint_Q f = \iint_Q f(x, y) dx dy = \sum_{j=1}^m \sum_{i=1}^n c_{ij} (x_i - x_{i-1}) (y_j - y_{j-1})$$

This is exactly the sum of the volumes of the rectangular blocks.

If f is a constant step function on the interior of Q , say $f(x, y) = k$, then $\iint_Q f = k(b-a)(d-c)$. Notice that calculating the volume block-by-block yields the exact same result as performing **iterated integration**:

$$\iint_Q f = \int_a^b \left(\int_c^d f(x, y) dy \right) dx = \int_c^d \left(\int_a^b f(x, y) dx \right) dy$$

Theorem : Properties of Step Function Integrals

For any constants $c_1, c_2 \in \mathbb{R}$ and step functions $f, g : Q \rightarrow \mathbb{R}$:

- **Linearity:** $\iint_Q (c_1 f + c_2 g) = c_1 \iint_Q f + c_2 \iint_Q g$
- **Additivity:** If Q is subdivided into rectangles Q_1 and Q_2 , then $\iint_Q f = \iint_{Q_1} f + \iint_{Q_2} f$
- **Comparison:** If $f(x, y) \leq g(x, y)$ for all $(x, y) \in Q$, then $\iint_Q f \leq \iint_Q g$.

35. Integrability of Bounded Functions

Let $f : Q \rightarrow \mathbb{R}$ be a **bounded function** on Q . This means $|f(x, y)| \leq M$ for all $(x, y) \in Q$. We can always "trap" f between two step functions s and t such that:

$$s(x, y) \leq f(x, y) \leq t(x, y) \quad \forall (x, y) \in Q$$

For example, the trivial step functions $s(x, y) = -M$ and $t(x, y) = M$ work perfectly.

Definition : Upper and Lower Integrals

Assume $f : Q \rightarrow \mathbb{R}$ is bounded. Let s and t be step functions such that $s \leq f \leq t$. Define two non-empty sets of real numbers:

$$S = \left\{ \iint_Q s \mid s \text{ is a step function with } s \leq f \right\}$$

$$T = \left\{ \iint_Q t \mid t \text{ is a step function with } t \geq f \right\}$$

Every number in T is an upper bound of S , and every number in S is a lower bound of T . Hence, the supremum of S and the infimum of T must exist.

- $\underline{I}(f) = \sup(S)$ is called the **lower integral** of f .
- $\bar{I}(f) = \inf(T)$ is called the **upper integral** of f .

Theorem : Integrability

For any bounded function f on Q , we have $\iint_Q s \leq \underline{I}(f) \leq \bar{I}(f) \leq \iint_Q t$. A bounded function f is said to be **integrable** over Q if and only if its lower and upper integrals are equal:

$$\underline{I}(f) = \bar{I}(f)$$

In this case, the unique real number $\underline{I}(f) = \bar{I}(f)$ is denoted by $\iint_Q f$.

36. Integrability of Continuous Functions

Not all bounded functions are integrable. However, all *continuous* functions on a closed rectangle are.

Theorem : Continuity Implies Small Span

Let $f : [a_1, b_1] \times \cdots \times [a_n, b_n] \rightarrow \mathbb{R}$ be continuous. Then given any $\epsilon > 0$, there exists a partition of the rectangle into a finite number of subrectangles such that the **span of f** (defined as $\max(f) - \min(f)$) is strictly less than ϵ in every subrectangle.

Proof:

(Proof by Contradiction via Nested Rectangles)

Suppose the result is false. Then there exists an $\epsilon_0 > 0$ such that the rectangle cannot be partitioned into subrectangles where the span is $< \epsilon_0$.

Consider the partition formed by bisecting every interval $[a_i, b_i]$. This divides the main rectangle into 2^n smaller subrectangles. If the theorem is false, at least one of these subrectangles (call it $I^{(1)}$) must still have a span $\geq \epsilon_0$.

We bisect $I^{(1)}$ to get a smaller "bad" subrectangle $I^{(2)} \subseteq I^{(1)}$ where the span is still $\geq \epsilon_0$. Continuing this indefinitely yields a nested chain $I^{(1)} \supseteq I^{(2)} \supseteq \cdots \supseteq I^{(m)} \dots$

By the completeness of the real numbers (Nested Interval Theorem), there is a unique point $t = (t_1, \dots, t_n)$ which lies in all these subrectangles. Because f is continuous at t , there exists an open ball $B(t; \delta)$ such that the span of f in that ball is $< \frac{1}{2}\epsilon_0$.

However, for a sufficiently large m , the subrectangle $I^{(m)}$ becomes so small that it fits entirely inside $B(t; \delta)$. This implies the span of f on $I^{(m)}$ must be $< \frac{1}{2}\epsilon_0$, which is a direct **contradiction** to our assumption that every $I^{(m)}$ has a span $\geq \epsilon_0$. The theorem holds. ■

37. Fubini's Theorem for Continuous Functions

Theorem : Integrability and Iterated Integrals

If $f : Q \rightarrow \mathbb{R}$ (where $Q = [a, b] \times [c, d]$) is continuous, then f is integrable on Q . Moreover, the double integral exactly equals the iterated integrals:

$$\iint_Q f = \int_a^b \left(\int_c^d f(x, y) dy \right) dx = \int_c^d \left(\int_a^b f(x, y) dx \right) dy$$

Proof:

1. Proving Integrability:

Since f is continuous on a closed rectangle, it is bounded, so it has upper ($\bar{I}$) and lower ($\underline{I}$) integrals. We must prove $\bar{I}(f) = \underline{I}(f)$.

Let $\epsilon > 0$. By the previous theorem, there is a partition P dividing Q into subrectangles $Q_1, \dots, Q_d$ such that the span of f on each Q_i is $< \epsilon$. Let $M_i(f)$ and $m_i(f)$ be the max and min values of f on Q_i . Thus $M_i(f) - m_i(f) < \epsilon$.

Define a step function t using the max values ($t = M_i$) and s using the min values ($s = m_i$). Then $s \leq f \leq t$ on Q . The difference between their integrals is:

$$\iint_Q t - \iint_Q s = \sum_{i=1}^d (M_i(f) - m_i(f)) \text{area}(Q_i) < \epsilon \sum_{i=1}^d \text{area}(Q_i) = \epsilon \cdot \text{area}(Q)$$

Because $\iint s \leq \underline{I}(f) \leq \bar{I}(f) \leq \iint t$, the difference $\bar{I}(f) - \underline{I}(f) < \epsilon \cdot \text{area}(Q)$. Letting $\epsilon \rightarrow 0$ proves $\bar{I}(f) = \underline{I}(f)$, hence f is integrable.

2. Proving Equality to Iterated Integrals:

Let $A(y) = \int_a^b f(x, y) dx$. We claim $A(y)$ is a continuous function of y on $[c, d]$.

$$|A(y) - A(y_1)| \leq \int_a^b |f(x, y) - f(x, y_1)| dx \leq (b - a) \max_{x \in [a, b]} |f(x, y) - f(x, y_1)|$$

Because f is continuous, as $y \rightarrow y_1$, the max difference goes to 0, hence $A(y) \rightarrow A(y_1)$, proving $A(y)$ is continuous. Because $A(y)$ is continuous, its integral over $[c, d]$ exists. We conclude:

$$\iint_Q f = \int_c^d A(y) dy = \int_c^d \left(\int_a^b f(x, y) dx \right) dy$$

A symmetric argument holds for integrating dx first. ■

38. Sets of Content Zero and Integrability

To integrate over general regions, we must understand when discontinuities ruin an integral and when they can be safely ignored.

Definition : Content Zero

A bounded subset $A \subseteq \mathbb{R}^2$ is said to have **content zero** if for any $\epsilon > 0$, there exists a finite set of rectangles $R_1, \dots, R_n$ in $\mathbb{R}^2$ such that:

$$A \subseteq \bigcup_{i=1}^n R_i \quad \text{and} \quad \sum_{i=1}^n \text{area}(R_i) < \epsilon$$

Examples of sets with content zero:

1. Any finite set of points in $\mathbb{R}^2$.
2. The union of a finite number of bounded sets of content zero.
3. Any subset of a bounded set of content zero.
4. Any line segment or the graph of a continuous function.

Theorem : Integrability over Discontinuities

Let $f : Q \rightarrow \mathbb{R}$ be a bounded function on the rectangle $Q = [a, b] \times [c, d]$. Let $D(f)$ be the set of discontinuities of f in Q :

$$D(f) = \{(x, y) \in Q \mid f \text{ is not continuous at } (x, y)\}$$

If $D(f)$ has content zero, then f is integrable over Q .

Proof:

Since f is bounded on Q , let $M > 0$ be such that $|f(x, y)| \leq M$ for all $(x, y) \in Q$. Given $\delta > 0$, because $D(f)$ has content zero, let P be a partition of Q such that the sum of the areas of all subrectangles of P which contain points of $D(f)$ is strictly less than δ . Now, given $\epsilon > 0$, refine P if necessary to get a partition of Q such that the span of f on the subrectangles which do **not** intersect $D(f)$ is less than ϵ . (This is possible because f is uniformly continuous on the closed subset of Q excluding the δ -neighborhoods of $D(f)$).

Define two step functions s and t :

- On subrectangles intersecting $D(f)$: set $s = -M$ and $t = M$.
- On subrectangles NOT intersecting $D(f)$: set $s = \min(f)$ and $t = \max(f)$ for that specific rectangle.

By definition, $s \leq f \leq t$ on Q . Now evaluate the difference of their integrals:

$$\iint_Q t - \iint_Q s \leq \epsilon \cdot \text{area}(Q) + 2M\delta$$

Because the upper and lower integrals satisfy $0 \leq \bar{I}(f) - \underline{I}(f) \leq \iint_Q t - \iint_Q s$, we have:

$$0 \leq \bar{I}(f) - \underline{I}(f) \leq \epsilon \cdot \text{area}(Q) + 2M\delta$$

Since ϵ and δ are arbitrary and can be chosen to be infinitesimally small, we must have $\underline{I}(f) = \overline{I}(f)$. Thus, f is integrable over Q . ■

39. Integration over Non-Rectangular Regions

Let $S \subseteq \mathbb{R}^2$ be a bounded region enclosed in a rectangle $Q \subseteq \mathbb{R}^2$. Let $f : S \rightarrow \mathbb{R}$ be a bounded function. We define $\tilde{f} : Q \rightarrow \mathbb{R}$ by extending f with zeros outside of S :

$$\tilde{f}(x, y) = \begin{cases} f(x, y) & \text{if } (x, y) \in S \\ 0 & \text{if } (x, y) \in Q \setminus S \end{cases}$$

Definition: f is said to be integrable over S if $\tilde{f}$ is integrable over Q , and we define $\iint_S f = \iint_Q \tilde{f}$.

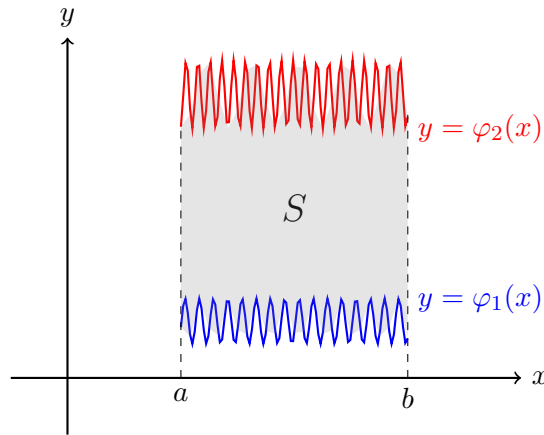
Definition : Type I and Type II Regions

1. Type I Region: A bounded region S is of Type I if it is bounded by two vertical lines and two continuous functions of x :

$$S = \{(x, y) \mid a \leq x \leq b, \varphi_1(x) \leq y \leq \varphi_2(x)\}$$

2. Type II Region: A bounded region T is of Type II if it is bounded by two horizontal lines and two continuous functions of y :

$$T = \{(x, y) \mid c \leq y \leq d, \psi_1(y) \leq x \leq \psi_2(y)\}$$



Theorem : Iterated Integrals on Type I and II Regions

Let S be a region of Type I. Suppose $f : S \rightarrow \mathbb{R}$ is a bounded function and continuous on the interior of S . Then f is integrable over S , and:

$$\iint_S f = \int_a^b \left(\int_{\varphi_1(x)}^{\varphi_2(x)} f(x, y) dy \right) dx$$

Similarly, for a Type II region T , $\iint_T f = \int_c^d \left(\int_{\psi_1(y)}^{\psi_2(y)} f(x, y) dx \right) dy$.

If a region is both Type I and Type II (like a circle or ellipse), both iterated integrals yield the identical volume.

40. Application: Volume of an Ellipsoid

Computing the Volume enclosed by an Ellipsoid

Compute the volume V of the solid enclosed by the ellipsoid:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Solution:

The upper surface is $z = c\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}$ (let's call this $g(x, y)$), and the lower surface is $z = -g(x, y)$ (let's call this $f(x, y)$). The solid lies over the 2D elliptical region $S = \{(x, y) \in \mathbb{R}^2 \mid \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1\}$.

By symmetry, the total volume V is 8 times the volume in the first octant ($x \geq 0, y \geq 0, z \geq 0$). In the first octant, x ranges from 0 to a . For a fixed x , y ranges from 0 to $b\sqrt{1 - \frac{x^2}{a^2}}$.

$$V = 8 \int_0^a \left(\int_0^{b\sqrt{1-x^2/a^2}} c\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} dy \right) dx$$

Let $A = \sqrt{1 - \frac{x^2}{a^2}}$. The inner integral becomes:

$$\int_0^{bA} c\sqrt{A^2 - \frac{y^2}{b^2}} dy$$

We use the trigonometric substitution $y = Ab \sin t \implies dy = Ab \cos t dt$. When $y = 0, t = 0$. When $y = bA, t = \pi/2$.

$$\begin{aligned} \int_0^{\pi/2} c\sqrt{A^2 - A^2 \sin^2 t} (Ab \cos t) dt &= \int_0^{\pi/2} c(A \cos t)(Ab \cos t) dt \\ &= cA^2 b \int_0^{\pi/2} \cos^2 t dt = cA^2 b \left(\frac{\pi}{4} \right) \end{aligned}$$

Substitute this result back into the outer integral (remembering $A^2 = 1 - \frac{x^2}{a^2}$):

$$\begin{aligned} V &= 8 \int_0^a c \left(1 - \frac{x^2}{a^2} \right) b \left(\frac{\pi}{4} \right) dx = 2\pi bc \int_0^a \left(1 - \frac{x^2}{a^2} \right) dx \\ &= 2\pi bc \left[x - \frac{x^3}{3a^2} \right]_0^a = 2\pi bc \left(a - \frac{a}{3} \right) = 2\pi bc \left(\frac{2a}{3} \right) = \frac{4}{3} \pi abc \end{aligned}$$

If $a = b = c$, this perfectly reduces to the volume of a sphere: $\frac{4}{3}\pi a^3$.

41. Appendix: Continuity of the Iterated Integral

During the proof for Type I regions, we assumed that the function $B(x) = \int_{\varphi_1(x)}^{\varphi_2(x)} f(x, y) dy$ is continuous. We prove this rigorously here.

Proof:

Let $x_0 \in [a, b]$. Let $\alpha = \varphi_1(x_0)$ and $\beta = \varphi_2(x_0)$. Since φ_1, φ_2 are continuous at x_0 , for a given $\rho > 0$, there exists $\delta > 0$ such that:

$$|\varphi_2(x) - \varphi_2(x_0)| < \rho \quad \text{and} \quad |\varphi_1(x) - \varphi_1(x_0)| < \rho$$

whenever $|x - x_0| < \delta$. Thus, $\varphi_1(x)$ stays near α , and $\varphi_2(x)$ stays near β .

Since f is continuous on the compact set S , f is uniformly continuous on S . Let M_2 be a global bound for the difference in y -values of the curves. We can choose $\delta > 0$ small enough such that:

$$|f(x, y) - f(x_0, y)| < \frac{\epsilon}{3M_2}$$

Evaluating the difference $|B(x) - B(x_0)|$:

$$\begin{aligned} |B(x) - B(x_0)| &= \left| \int_{\varphi_1(x)}^{\varphi_2(x)} f(x, y) dy - \int_{\varphi_1(x_0)}^{\varphi_2(x_0)} f(x_0, y) dy \right| \\ &\leq \left| \int_{\varphi_2(x_0)}^{\varphi_2(x)} f(x, y) dy \right| \\ &\quad + \int_{\varphi_1(x_0)}^{\varphi_2(x_0)} |f(x, y) - f(x_0, y)| dy \quad + \left| \int_{\varphi_1(x)}^{\varphi_1(x_0)} f(x, y) dy \right| \end{aligned}$$

The middle integral is bounded by the uniform continuity of f : $(\beta - \alpha) \frac{\epsilon}{3M_2} \leq M_2 \frac{\epsilon}{3M_2} = \frac{\epsilon}{3}$. The first and third integrals are integrals over very tiny slivers of width $< \rho$. Because f is bounded by some global maximum M_1 , these outer integrals are bounded by $M_1\rho$. By choosing ρ sufficiently small, we can bound the entire sum by ϵ , proving $B(x)$ is continuous at x_0 . ■

42. Evaluating Complex Double Integrals

When evaluating integrals involving absolute values, we must carefully partition the region of integration to remove the absolute value signs.

Example 1: Absolute Value over a Rectangle

Evaluate $\iint_Q \sqrt{|y - x^2|} dx dy$ over $Q = [-1, 1] \times [0, 2]$.

Solution: The curve $y = x^2$ divides the rectangle into two distinct sub-regions. Inside Q , if $y > x^2$, then $|y - x^2| = y - x^2$. If $y < x^2$, then $|y - x^2| = x^2 - y$.

We fix x and integrate with respect to y first:

$$B(x) = \int_0^2 \sqrt{|y - x^2|} dy = \int_0^{x^2} \sqrt{x^2 - y} dy + \int_{x^2}^2 \sqrt{y - x^2} dy$$

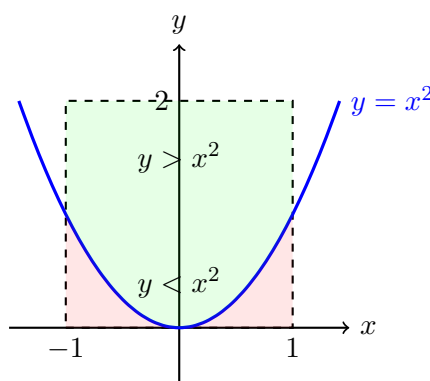
Using the substitutions $t = x^2 - y$ (for the first) and $t = y - x^2$ (for the second):

$$B(x) = \int_{x^2}^0 -\sqrt{t} dt + \int_0^{2-x^2} \sqrt{t} dt = \int_0^{x^2} t^{1/2} dt + \int_0^{2-x^2} t^{1/2} dt$$

$$B(x) = \frac{2}{3}(x^2)^{3/2} + \frac{2}{3}(2-x^2)^{3/2} = \frac{2}{3}|x|^3 + \frac{2}{3}(2-x^2)^{3/2}$$

Since we are integrating x from -1 to 1 , x^3 is odd, so we carefully integrate the remaining symmetric terms:

$$\iint_Q \sqrt{|y-x^2|} dx dy = \int_{-1}^1 \left[\frac{2}{3}|x|^3 + \frac{2}{3}(2-x^2)^{3/2} \right] dx = \frac{1}{3} + \frac{4}{3} \int_0^1 (2-x^2)^{3/2} dx$$



43. Change of Variables and the Jacobian

To integrate over complex domains, we map a simple region T to a complex region S using a transformation g . The **Jacobian determinant** acts as a scaling factor for the area/volume.

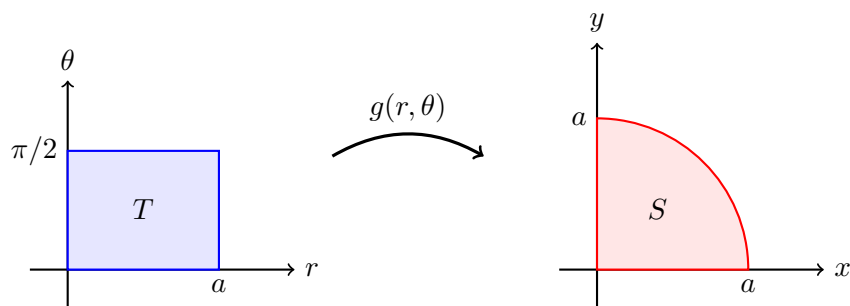
Theorem : Change of Variables in 2D

Let $T \xrightarrow{g} S \xrightarrow{f} \mathbb{R}$, where $g(u, v) = (x(u, v), y(u, v))$. We define the Jacobian determinant $J(u, v)$ as:

$$J(u, v) = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = \frac{\partial(x, y)}{\partial(u, v)}$$

Assuming g is 1-1 on the interior of T , f is continuous, and the partials exist and are continuous:

$$\iint_S f(x, y) dx dy = \iint_T f(x(u, v), y(u, v)) |J(u, v)| du dv$$



Classic Coordinate Transformations

1. Polar Coordinates: $g(r, \theta) = (r \cos \theta, r \sin \theta)$.

$$J(r, \theta) = \det \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix} = r \cos^2 \theta + r \sin^2 \theta = r$$

2. Cylindrical Coordinates: $g(r, \theta, z) = (r \cos \theta, r \sin \theta, z)$.

$$J(r, \theta, z) = \det \begin{pmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = r$$

3. Spherical Coordinates: $g(\rho, \theta, \phi) = (\rho \cos \theta \sin \phi, \rho \sin \theta \sin \phi, \rho \cos \phi)$.

$$J(\rho, \theta, \phi) = -\rho^2 \sin \phi \implies |J| = \rho^2 \sin \phi$$

44. Volume of the n -Dimensional Sphere

Let $S_n(a) = \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_1^2 + \dots + x_n^2 \leq a^2\}$ be the n -dimensional sphere of radius a . Its volume is denoted $V_n(a) = \int \dots \int_{S_n(a)} dx_1 \dots dx_n$. By factoring out a^n , we have $V_n(a) = a^n V_n(1)$.

Theorem : Recursive Formula for $V_n(1)$

The volume of the unit n -sphere satisfies the recurrence relation:

$$V_n(1) = \frac{2\pi}{n} V_{n-2}(1) \quad \text{for } n \geq 3$$

Proof:

We isolate the last two variables, x_{n-1} and x_n . The condition $x_1^2 + \dots + x_n^2 \leq 1$ implies that the first $n-2$ variables form an $(n-2)$ -dimensional sphere of radius $\sqrt{1 - x_{n-1}^2 - x_n^2}$.

$$\begin{aligned} V_n(1) &= \iint_{x_{n-1}^2 + x_n^2 \leq 1} \left(\int \dots \int_{x_1^2 + \dots + x_{n-2}^2 \leq 1 - x_{n-1}^2 - x_n^2} dx_1 \dots dx_{n-2} \right) dx_{n-1} dx_n \\ &= \iint_{x_{n-1}^2 + x_n^2 \leq 1} V_{n-2} \left(\sqrt{1 - x_{n-1}^2 - x_n^2} \right) dx_{n-1} dx_n \\ &= \iint_{x_{n-1}^2 + x_n^2 \leq 1} (1 - x_{n-1}^2 - x_n^2)^{\frac{n-2}{2}} V_{n-2}(1) dx_{n-1} dx_n \end{aligned}$$

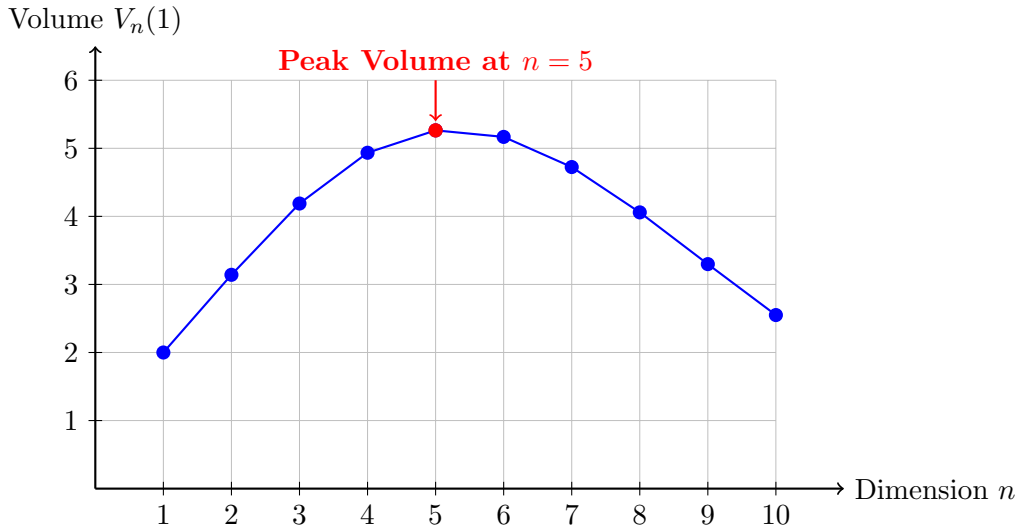
To evaluate this double integral, we switch to polar coordinates: $x_{n-1} = r \cos \theta, x_n = r \sin \theta$, with Jacobian r .

$$\begin{aligned} V_n(1) &= V_{n-2}(1) \int_0^{2\pi} \int_0^1 (1 - r^2)^{\frac{n-2}{2}} r dr d\theta \\ &= V_{n-2}(1) (2\pi) \left[\frac{-(1 - r^2)^{n/2}}{n} \right]_0^1 = \frac{2\pi}{n} V_{n-2}(1) \end{aligned}$$

■

Using the base cases $V_1(1) = 2$ (length of interval $[-1, 1]$) and $V_2(1) = \pi$ (area of unit circle), we find:

$$V_3(1) = \frac{4\pi}{3}, \quad V_4(1) = \frac{\pi^2}{2}, \quad V_5(1) = \frac{8\pi^2}{15}, \quad V_6(1) = \frac{\pi^3}{6}$$



The Gamma Function Formulation

Using the Gamma function $\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt$, where $\Gamma(s+1) = s\Gamma(s)$, $\Gamma(1) = 1$, and $\Gamma(1/2) = \sqrt{\pi}$, we can express the volume cleanly without recursion.

Let $f(n) = \frac{\pi^{n/2}}{\Gamma(n/2+1)}$. By observing that $\frac{f(n-2)}{f(n)}$ perfectly mirrors our recurrence relation $\frac{n}{2\pi}$, and verifying the base cases, we conclude the universal formula for the volume of a unit n -sphere is:

$$V_n(1) = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)}$$

45. Infinite Series of Real Numbers

Reference: Ross, Elementary Analysis (May 4th, 2026)

Definitions and Limits

Recall that for a sequence of real numbers (s_n) , let S be the set of all limits of convergent subsequences of (s_n) .

$$\limsup_{n \rightarrow \infty} s_n = \lim_{N \rightarrow \infty} \sup\{s_n \mid n \geq N\} = \sup(S)$$

$$\liminf_{n \rightarrow \infty} s_n = \lim_{N \rightarrow \infty} \inf\{s_n \mid n \geq N\} = \inf(S)$$

Definition : Convergence of a Series

Consider an infinite series of real numbers $\sum_{n=m}^\infty a_n$, where m is some starting integer (often 0 or 1). We define the n -th **partial sum** of the series as:

$$s_n = \sum_{i=m}^n a_i = a_m + a_{m+1} + \cdots + a_n \quad \text{for } n \geq m$$

The series $\sum_{n=m}^\infty a_n$ is said to **converge** to a real number s if the sequence of partial sums converges to s , i.e., $\lim_{n \rightarrow \infty} s_n = s$. In this case, we write:

$$\sum_{n=m}^\infty a_n = s$$

If $\lim_{n \rightarrow \infty} s_n = +\infty$, we say the series **diverges to** $+\infty$.

If $\lim_{n \rightarrow \infty} s_n = -\infty$, we say the series **diverges to** $-\infty$.

If $\lim_{n \rightarrow \infty} s_n$ does not exist (neither as a real number nor $\pm\infty$), we simply say the series **diverges**.

Examples of Divergence and Geometric Series

1. Simple Divergence: Let $a_n = 1$. Then $\sum_{n=1}^\infty a_n = \infty$. Let $a_n = -1$. Then $\sum_{n=1}^\infty a_n = -\infty$.

2. Oscillating Divergence: Let $a_n = (-1)^{n+1}$. The series is $1 - 1 + 1 - 1 \dots$. The partial sums alternate between 1 and 0. Since the limit does not exist, the series diverges.

3. Geometric Series: Let $a_n = ar^n$ for fixed real numbers a, r . The series is $\sum_{n=0}^{\infty} ar^n$. If $r \neq 1$, the partial sum is $s_n = a \frac{1-r^{n+1}}{1-r}$. If $|r| < 1$, then $r^{n+1} \rightarrow 0$ as $n \rightarrow \infty$, so $\lim_{n \rightarrow \infty} s_n = \frac{a}{1-r}$. Thus, for $|r| < 1$, the geometric series converges to $\frac{a}{1-r}$.

Definition : Absolute Convergence

A series $\sum a_n$ **converges absolutely** if the series of absolute values $\sum |a_n|$ is a convergent series.

46. The Cauchy Criterion for Series

Theorem : Cauchy Condition

A series of real numbers $\sum a_n$ satisfies the **Cauchy condition** if its sequence of partial sums (s_n) is a Cauchy sequence.

Formally: For any $\epsilon > 0$, there exists an integer $N > 0$ such that for all $n \geq m > N$, we have $|s_n - s_{m-1}| < \epsilon$. This is equivalent to:

$$|a_m + a_{m+1} + \cdots + a_n| < \epsilon$$

Because $\mathbb{R}$ is complete, a series converges if and only if it satisfies the Cauchy condition.

Theorem: Absolute Convergence Implies Convergence

Proposition: $\sum a_n$ converges absolutely $\implies \sum a_n$ converges.

Proof: Assume $\sum a_n$ converges absolutely. This means $\sum |a_n|$ converges, and thus satisfies the Cauchy condition. Given $\epsilon > 0$, there exists an N such that for all $n \geq m > N$:

$$||a_m| + |a_{m+1}| + \cdots + |a_n|| < \epsilon$$

Since absolute values are non-negative, this is simply:

$$|a_m| + |a_{m+1}| + \cdots + |a_n| < \epsilon$$

By the Triangle Inequality:

$$|a_m + a_{m+1} + \cdots + a_n| \leq |a_m| + |a_{m+1}| + \cdots + |a_n| < \epsilon$$

Thus, $\sum a_n$ satisfies the Cauchy condition, and therefore converges. ■

Corollary: The n -th Term Test

Proposition: If $\sum a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$.

Proof: If $\sum a_n$ converges, it satisfies the Cauchy condition. Given $\epsilon > 0$, there exists N such that for all $n \geq m > N$, $|a_m + \cdots + a_n| < \epsilon$. Choose $n = m$. Then $|a_n| < \epsilon$ for all $n > N$. This is the exact definition of $\lim_{n \rightarrow \infty} a_n = 0$. ■

47. Convergence Tests for Non-Negative Series

For this section, assume we are dealing with series $\sum a_n$ where $a_n \geq 0$ for all n . The sequence of partial sums s_n is strictly non-decreasing. Therefore, s_n either converges to a finite supremum or diverges to $+\infty$.

Theorem : The Comparison Test

1. If $\sum a_n$ converges and $|b_n| \leq a_n$ for all n , then $\sum b_n$ converges.
2. If $\sum a_n$ diverges to $+\infty$ and $b_n \geq a_n \geq 0$ for all n , then $\sum b_n$ diverges to $+\infty$.

Theorem : The Root Test

Let $\sum a_n$ be a series of reals. Let $\alpha = \limsup_{n \rightarrow \infty} |a_n|^{1/n}$.

1. If $\alpha < 1$, the series converges absolutely.
2. If $\alpha > 1$, the series diverges.
3. If $\alpha = 1$, the test yields no information.

Proof:

Proof of (1): Let $\alpha < 1$. Choose $\epsilon > 0$ such that $\alpha + \epsilon < 1$. By the definition of $\limsup$, there exists an integer N such that $|a_n|^{1/n} < \alpha + \epsilon$ for all $n > N$. This implies $|a_n| < (\alpha + \epsilon)^n$ for all $n > N$. Because $\alpha + \epsilon < 1$, the geometric series $\sum (\alpha + \epsilon)^n$ converges. By the Comparison Test, $\sum |a_n|$ converges.

Proof of (2): Let $\alpha > 1$. Then there exists a subsequence of $|a_n|^{1/n}$ that converges to α . This means there are infinitely many n such that $|a_n|^{1/n} > 1$, which implies $|a_n| > 1$. Since the terms do not approach 0 ($\lim a_n \neq 0$), the series must diverge by the n -th term test. ■

48. The p -Series and The Integral Test

We analyze the convergence of the p -series: $\sum_{n=1}^{\infty} \frac{1}{n^p}$.

Case 1: $p = 2$

We want to evaluate $\sum_{n=1}^{\infty} \frac{1}{n^2}$. Notice that for $n \geq 2$:

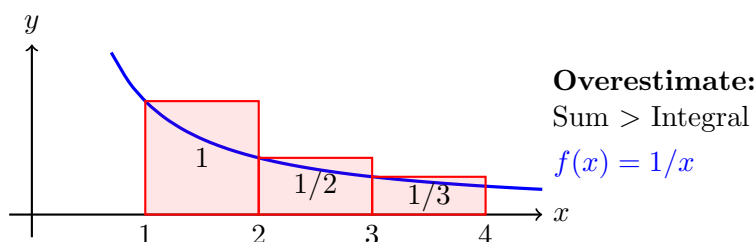
$$\frac{1}{n^2} \leq \frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}$$

The partial sum s_n is bounded:

$$s_n = 1 + \frac{1}{2^2} + \cdots + \frac{1}{n^2} \leq 1 + \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n-1} - \frac{1}{n}\right) = 2 - \frac{1}{n}$$

Since $s_n \leq 2$ for all n and s_n is non-decreasing, the sequence converges (and its limit is ≤ 2 , famously $\pi^2/6$).

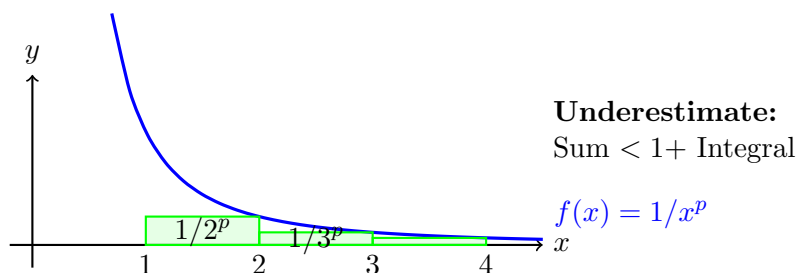
Case 2: The Integral Test for General p



Divergence of Harmonic Series ($p = 1$): By drawing rectangles of width 1 and height $f(n) = 1/n$ anchored on the left, the area of the rectangles completely covers the area under the curve $f(x) = 1/x$. Thus:

$$\sum_{k=1}^n \frac{1}{k} \geq \int_1^{n+1} \frac{1}{x} dx = \log(n+1)$$

As $n \rightarrow \infty$, $\log(n+1) \rightarrow \infty$. Thus the harmonic series diverges to $+\infty$. By the comparison test, any $\sum \frac{1}{n^p}$ for $p \leq 1$ also diverges.



Convergence for $p > 1$: By shifting the rectangles to be anchored on the right (starting from $x = 1$), the rectangles fit entirely under the curve. We separate the first term:

$$s_n = 1 + \frac{1}{2^p} + \frac{1}{3^p} + \cdots + \frac{1}{n^p} \leq 1 + \int_1^n \frac{1}{x^p} dx$$

Evaluating the integral:

$$\int_1^n x^{-p} dx = \left[\frac{x^{-(p-1)}}{-(p-1)} \right]_1^n = \frac{1}{p-1} - \frac{1}{(p-1)n^{p-1}}$$

As $n \rightarrow \infty$, the term $\frac{1}{n^{p-1}} \rightarrow 0$ (because $p-1 > 0$). Thus, the integral is bounded by $\frac{1}{p-1}$. The partial sums are bounded: $s_n \leq 1 + \frac{1}{p-1} = \frac{p}{p-1}$. Since s_n is a bounded, non-decreasing sequence, it converges.

Conclusion: The p -series $\sum \frac{1}{n^p}$ converges $\iff p > 1$.

49. The Ratio Test and Its Limits

Theorem : The Ratio Test

Let $\sum a_n$ be a series with non-zero terms ($a_n \neq 0$ for all n).

1. $\sum a_n$ **converges absolutely** if $\limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$.
2. $\sum a_n$ **diverges** if $\liminf_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| > 1$.
3. Otherwise, $\liminf_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \leq 1 \leq \limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$, and the **test fails** (yields no information).

Examples of Failure: For $a_n = \frac{1}{n}$, the ratio is $\frac{n}{n+1} \rightarrow 1$. For $b_n = \frac{1}{n^2}$, the ratio is $\frac{n^2}{(n+1)^2} \rightarrow 1$. Both yield a limit of 1, but $\sum \frac{1}{n}$ diverges while $\sum \frac{1}{n^2}$ converges.

Theorem : Ratio vs. Root Test Bounds

Let (a_n) be a sequence of strictly non-zero real numbers. Then the limit superiors and limit inferiors of the root and ratio tests obey the following strict inequality chain:

$$\liminf_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \leq \liminf_{n \rightarrow \infty} |a_n|^{1/n} \leq \limsup_{n \rightarrow \infty} |a_n|^{1/n} \leq \limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

(The middle inequality is obvious by definition. The outer inequalities require formal proof.)

Conclusion: The Root Test is strictly stronger than the Ratio Test. If the Ratio Test gives a definitive answer, the Root Test will too. But the Root Test can succeed where the Ratio Test fails.

50. Examples of Test Application

Example 1: Where Ratio and Root Tests Both Fail

Analyze the convergence of the series: $\sum_{n=1}^{\infty} \frac{n}{n^2+3}$

Attempting the Ratio Test: Let $a_n = \frac{n}{n^2+3}$.

$$\frac{a_{n+1}}{a_n} = \frac{n+1}{(n+1)^2+3} \cdot \frac{n^2+3}{n} = \frac{n+1}{n} \cdot \frac{n^2+3}{n^2+2n+4}$$

Dividing numerator and denominator by the highest powers of n :

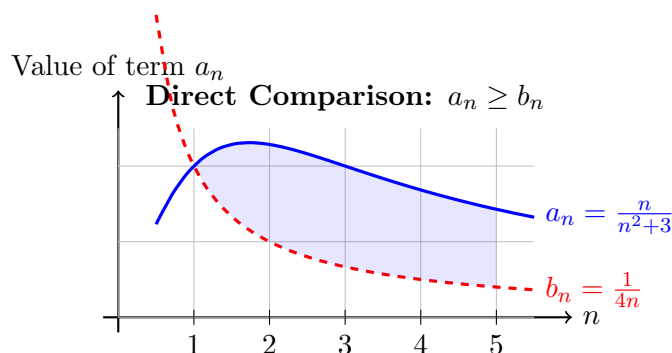
$$\frac{a_{n+1}}{a_n} = \left(\frac{1+1/n}{1} \right) \cdot \left(\frac{1+3/n^2}{1+2/n+4/n^2} \right)$$

As $n \rightarrow \infty$, this limit approaches $1 \cdot 1 = 1$. Both the Ratio Test and Root Test **fail**.

Using the Direct Comparison Test instead: Notice that for $n \geq 1$, $3 \leq 3n^2$. Therefore, $n^2+3 \leq n^2+3n^2 = 4n^2$. This means the fraction gets smaller if we make the denominator larger:

$$\frac{n}{n^2+3} \geq \frac{n}{n^2+3n^2} = \frac{n}{4n^2} = \frac{1}{4n}$$

We know that $\sum \frac{1}{n} = +\infty$ (Harmonic series diverges). Therefore, $\sum \frac{1}{4n} = +\infty$. Since our series is strictly greater than a diverging series, $\sum \frac{n}{n^2+3}$ **diverges to** $+\infty$.



Example 2: Ratio Test Fails, Root Test Succeeds

Analyze the convergence of the series: $\sum_{n=0}^{\infty} 2^{(-1)^n - n}$. Writing out the first few terms: $2 + \frac{1}{4} + \frac{1}{2} + \frac{1}{16} + \frac{1}{8} + \frac{1}{64} + \dots$. Notice that $a_n = 2^{(-1)^n - n} \leq \frac{1}{2^{n-1}}$ for all n . This immediately implies convergence by comparison, but let's test our specific convergence tools.

Attempting the Ratio Test:

$$\frac{a_{n+1}}{a_n} = \frac{2^{(-1)^{n+1} - (n+1)}}{2^{(-1)^n - n}} = 2^{(-1)^{n+1} - (-1)^n - 1}$$

We must split this into cases based on parity:

- If n is **even**: $(-1)^{n+1} = -1$ and $(-1)^n = 1$. The ratio is $2^{-1-1-1} = 2^{-3} = \frac{1}{8}$.
- If n is **odd**: $(-1)^{n+1} = 1$ and $(-1)^n = -1$. The ratio is $2^{1-(-1)-1} = 2^1 = 2$.

The sequence of ratios oscillates endlessly between $\frac{1}{8}$ and 2. Thus, $\liminf \left| \frac{a_{n+1}}{a_n} \right| = \frac{1}{8} \leq 1$ and $\limsup \left| \frac{a_{n+1}}{a_n} \right| = 2 \geq 1$. The **Ratio Test fails**.

Attempting the Root Test:

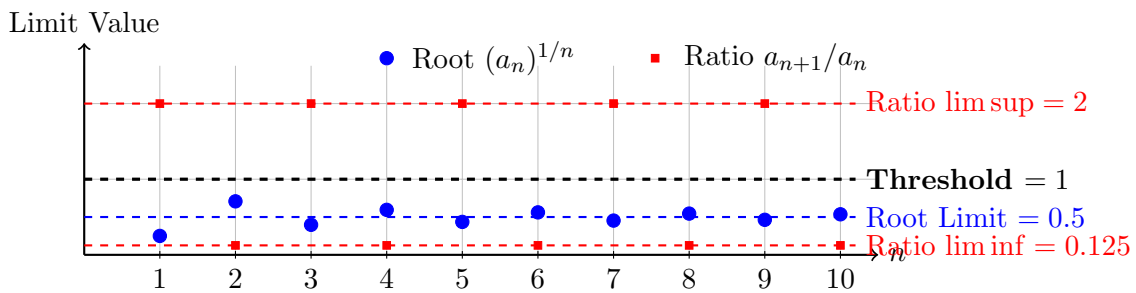
$$(a_n)^{1/n} = \left(2^{(-1)^n - n}\right)^{1/n} = 2^{\frac{(-1)^n}{n} - 1}$$

Again, evaluating parity limits as $n \rightarrow \infty$:

- If n is **even**: $(a_n)^{1/n} = 2^{1/n-1} \xrightarrow{n \rightarrow \infty} 2^{0-1} = \frac{1}{2}$.
- If n is **odd**: $(a_n)^{1/n} = 2^{-1/n-1} \xrightarrow{n \rightarrow \infty} 2^{0-1} = \frac{1}{2}$.

Therefore, $\limsup_{n \rightarrow \infty} |a_n|^{1/n} = \lim_{n \rightarrow \infty} |a_n|^{1/n} = \frac{1}{2}$.

Because $\frac{1}{2} < 1$, the **Root Test succeeds** and confirms the series converges absolutely.



51. The "Inconclusive" Boundary ($\alpha = 1$)

If $\alpha = \limsup |a_n|^{1/n} = 1$, then the Root Test yields no information. Consider the series $\sum \frac{1}{n}$ and $\sum \frac{1}{n^2}$. Both have $\alpha = 1$, but the first diverges while the second converges.

- If $a_n = \frac{1}{n}$, then $\limsup |a_n|^{1/n} = \lim_{n \rightarrow \infty} \left(\frac{1}{n}\right)^{1/n} = \frac{1}{\lim n^{1/n}} = 1$.
- If $a_n = \frac{1}{n^2}$, then $\limsup |a_n|^{1/n} = \lim_{n \rightarrow \infty} \left(\frac{1}{n^2}\right)^{1/n} = \frac{1}{(\lim n^{1/n})^2} = 1$.

52. The Ratio Test

Theorem : The Ratio Test

Let $\sum a_n$ be a series of nonzero terms ($a_n \neq 0$ for all n).

1. $\sum a_n$ **converges absolutely** if $\limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$.
2. $\sum a_n$ **diverges** if $\liminf_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| > 1$.
3. Else, if $\liminf \left| \frac{a_{n+1}}{a_n} \right| \leq 1 \leq \limsup \left| \frac{a_{n+1}}{a_n} \right|$, the test is **inconclusive**.

Note: If the standard limit $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ exists, then $\liminf = \limsup = \lim$. Hence, if this limit equals 1, the test is inconclusive.

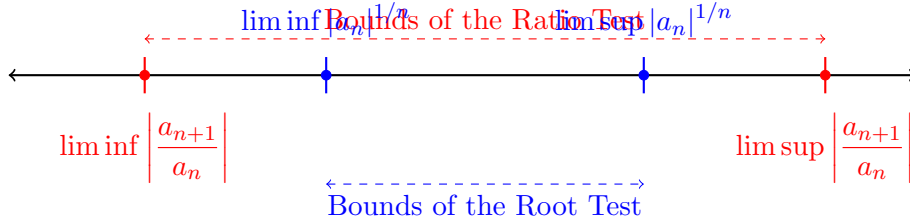
Proof: This test follows immediately from the Root Test and the Hierarchy Theorem below. For case (3), again, the series $\sum \frac{1}{n}$ and $\sum \frac{1}{n^2}$ both yield a limit of 1, showing the test is inconclusive.

53. The Grand Hierarchy: Ratio vs. Root Test

Theorem : Hierarchy of Convergence Tests

Let (a_n) be any sequence of nonzero real numbers. Then the limit superiors and inferiors of the ratios and roots satisfy the following nested inequalities:

$$\liminf_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \leq \liminf_{n \rightarrow \infty} |a_n|^{1/n} \leq \limsup_{n \rightarrow \infty} |a_n|^{1/n} \leq \limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$



Proof:

The middle inequality ($\liminf \leq \limsup$) is obvious by definition. We will prove the rightmost inequality:

$$\limsup_{n \rightarrow \infty} |a_n|^{1/n} \leq \limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

(The leftmost inequality is proved similarly).

Let $\alpha = \limsup_{n \rightarrow \infty} |a_n|^{1/n}$ and $L = \limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$.

We want to show that $\alpha \leq L$. If $L = +\infty$, the inequality is trivially true, so we may assume $L < \infty$. It is enough to prove that $\alpha \leq L_1$ for any arbitrary $L_1 > L$.

Because $L < L_1$, by the definition of the limit superior:

$$\lim_{N \rightarrow \infty} \left(\sup_{n \geq N} \left| \frac{a_{n+1}}{a_n} \right| \right) < L_1$$

Thus, there exists an integer $N > 0$ such that for all $n \geq N$, we have $\left| \frac{a_{n+1}}{a_n} \right| < L_1$.

For any $n > N$, we can write $|a_n|$ as a telescoping product:

$$|a_n| = |a_N| \cdot \left| \frac{a_{N+1}}{a_N} \right| \cdot \left| \frac{a_{N+2}}{a_{N+1}} \right| \cdots \left| \frac{a_n}{a_{n-1}} \right|$$

Since each of the $(n - N)$ fractions is strictly less than L_1 , we have:

$$|a_n| < |a_N| \cdot L_1^{n-N} = L_1^n (L_1^{-N} |a_N|) \quad \forall n \geq N$$

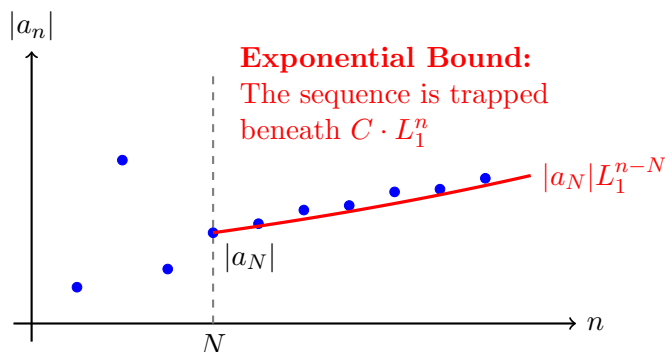
Let $a = L_1^{-N}|a_N|$, which is a constant. Then $|a_n| < L_1^n \cdot a$. Taking the n -th root of both sides:

$$|a_n|^{1/n} < L_1 \cdot a^{1/n} \quad \forall n \geq N$$

As $n \rightarrow \infty$, the constant root $a^{1/n} \rightarrow 1$. Therefore, taking the limit superior of both sides yields:

$$\alpha = \limsup_{n \rightarrow \infty} |a_n|^{1/n} \leq L_1 \cdot 1 = L_1$$

Since $\alpha \leq L_1$ for any $L_1 > L$, it must be that $\alpha \leq L$. This completes the proof. ■



54. Examples: Ratio vs. Root Test in Action

Example 1: Geometric Series

Evaluate $\sum_{n=2}^{\infty} \left(-\frac{1}{3}\right)^n = \frac{1}{9} - \frac{1}{27} + \frac{1}{81} - \frac{1}{243} + \dots$

This is a geometric series with $a = \frac{1}{9}$ and $r = -\frac{1}{3}$. Since $|r| < 1$, it converges:

$$\sum_{n=2}^{\infty} ar^n = \frac{a}{1-r} = \frac{1/9}{1-(-1/3)} = \frac{1/9}{4/3} = \frac{1}{12}$$

Example 2: Ratio & Root Tests Fail

Evaluate $\sum_{n=1}^{\infty} \frac{n}{n^2+3}$. Let $a_n = \frac{n}{n^2+3}$.

$$\frac{a_{n+1}}{a_n} = \frac{n+1}{(n+1)^2+3} \cdot \frac{n^2+3}{n} = \frac{n+1}{n} \cdot \frac{n^2+3}{n^2+2n+4}$$

Taking the limit as $n \rightarrow \infty$, $\lim \left| \frac{a_{n+1}}{a_n} \right| = 1$. Thus, the Root Test and Ratio Test give no information.

Instead, use the Comparison Test. Note that for $n \geq 1$:

$$\frac{n}{n^2+3} > \frac{n}{n^2+3n^2} = \frac{n}{4n^2} = \frac{1}{4n}$$

Since $\sum \frac{1}{n}$ diverges, $\sum \frac{1}{4n}$ diverges. Since our series is strictly greater, $\sum \frac{n}{n^2+3}$ **diverges**.

Example 3: Ratio/Root Fail, but Converges

Evaluate $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$. As before, $\frac{a_{n+1}}{a_n} \rightarrow 1$, so the Ratio and Root tests do not work. However, $\frac{1}{n^2+1} < \frac{1}{n^2}$ for all $n \geq 1$. Because we know the p -series $\sum \frac{1}{n^2}$ converges, our series **converges** by the Comparison Test.

Example 4: Ratio and Root Both Succeed

Evaluate $\sum_{n=1}^{\infty} \frac{n}{3^n}$. Let $a_n = \frac{n}{3^n}$. Ratio Test: $\frac{a_{n+1}}{a_n} = \frac{n+1}{3^{n+1}} \cdot \frac{3^n}{n} = \frac{n+1}{3n} \xrightarrow{n \rightarrow \infty} \frac{1}{3}$. Since $1/3 < 1$, it **converges**. Root Test: $|a_n|^{1/n} = \frac{n^{1/n}}{3} \xrightarrow{n \rightarrow \infty} \frac{1}{3}$. Both tests easily confirm convergence.

Example 5: Root Test Succeeds where Ratio is messy

Evaluate $\sum a_n$ where $a_n = \left(\frac{2}{(-1)^n - 3}\right)^n$.
Applying the Root Test:

$$|a_n|^{1/n} = \left| \frac{2}{(-1)^n - 3} \right| = \begin{cases} 1 & \text{for even } n \\ 1/2 & \text{for odd } n \end{cases}$$

The limit superior is $\limsup |a_n|^{1/n} = 1$. This implies the test might fail, BUT looking closer: for even n , $a_n = 1$. Since the terms do not tend to 0, the series **diverges** by the n -th term test.

Example 6: Ratio Fails, Root Succeeds

Evaluate $\sum_{n=0}^{\infty} 2^{(-1)^n - n} = 2 + \frac{1}{4} + \frac{1}{2} + \frac{1}{16} + \frac{1}{8} + \dots$

Let $a_n = 2^{(-1)^n - n}$. Note that $a_n \leq \frac{1}{2^{n-1}}$ for all n , so it converges by comparison. If we blindly apply the Ratio Test:

$$\frac{a_{n+1}}{a_n} = \frac{2^{(-1)^{n+1} - (n+1)}}{2^{(-1)^n - n}} = \begin{cases} 1/8 & \text{for even } n \\ 2 & \text{for odd } n \end{cases}$$

Thus $\liminf = 1/8 < 1 < \limsup = 2$. The Ratio Test is **inconclusive**.

However, applying the Root Test:

$$|a_n|^{1/n} = \begin{cases} 2^{1/n-1} & \text{for even } n \\ 2^{-1/n-1} & \text{for odd } n \end{cases}$$

In both cases, as $n \rightarrow \infty$, the limit approaches $2^{-1} = 1/2$. Since $\limsup = 1/2 < 1$, the Root Test definitively proves **absolute convergence**.

55. The Alternating Series Test

If $\lim |a_{n+1}/a_n| = 1$, exponential tests fail. If terms alternate signs, we use the Alternating Series Theorem.

Example 7: The Alternating p -series

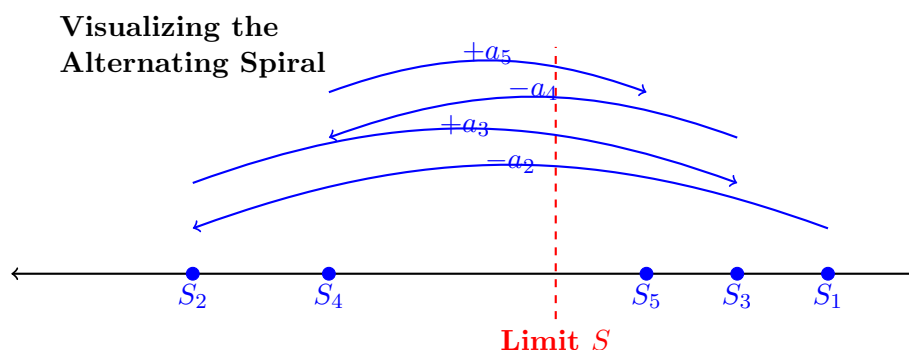
Evaluate $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}} = 1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{2} + \dots$

Applying the Ratio/Root tests yields $\lim = 1$ (inconclusive). However, it converges via the Alternating Series Test.

Theorem : Alternating Series Theorem

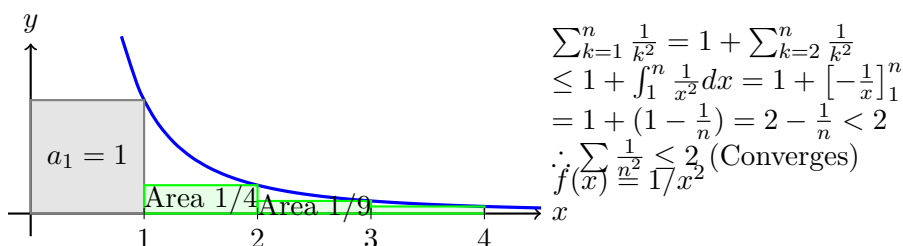
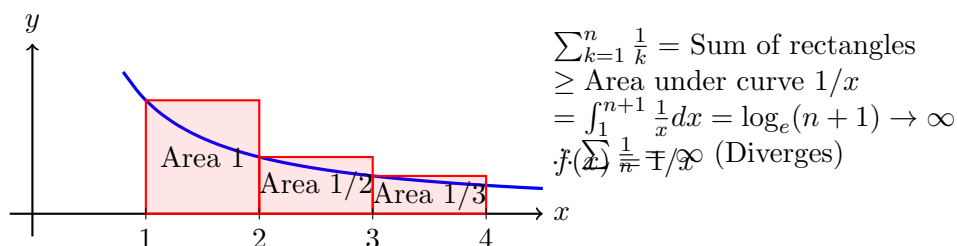
Let (a_n) be a sequence such that $a_1 \geq a_2 \geq \dots \geq a_n \geq 0$. If $\lim_{n \rightarrow \infty} a_n = 0$, then the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ converges to some limit S .

Moreover, the n -th partial sum $S_n = \sum_{k=1}^n (-1)^{k+1} a_k$ bounds the error of the sequence such that $|S - S_n| \leq a_{n+1}$ for all n .



56. The Integral Test and p -Series

If a series has strictly positive terms that decrease algebraically (like $\frac{1}{n}$ or $\frac{1}{n^2}$), we compare the discrete summation blocks to the continuous area under the curve.



Theorem : Convergence of the p -Series

The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if and only if $p > 1$.

Proof:

If $p > 1$:

By representing the sum as right-endpoint rectangles under the curve $f(x) = \frac{1}{x^p}$, we can bound the sum using the continuous integral:

$$\sum_{k=1}^n \frac{1}{k^p} \leq 1 + \int_1^n \frac{1}{x^p} dx = 1 + \left[\frac{x^{-(p-1)}}{-(p-1)} \right]_1^n = 1 + \frac{1}{p-1} \left(1 - \frac{1}{n^{p-1}} \right)$$

Because $p > 1$, the exponent $(p-1)$ is positive. As $n \rightarrow \infty$, the term $\frac{1}{n^{p-1}} \rightarrow 0$. Thus:

$$\sum_{n=1}^{\infty} \frac{1}{n^p} < 1 + \frac{1}{p-1} = \frac{p}{p-1} < \infty$$

Since the partial sums are strictly bounded above, the series converges.

If $0 < p \leq 1$:

For any $n \geq 1$, we have $n^p \leq n^1 = n$. Therefore, taking the reciprocal gives:

$$\frac{1}{n^p} \geq \frac{1}{n} \quad \forall n$$

We proved via the Integral Test that the harmonic series $\sum \frac{1}{n}$ diverges to ∞ . Since $\frac{1}{n^p}$ is strictly greater than or equal to the terms of a diverging series, $\sum \frac{1}{n^p}$ diverges by the Comparison Test. ■

57. Introduction to Power Series

Reference: Ross Chapter 4

Definition : Power Series

Consider the series $\sum_{n=0}^{\infty} a_n x^n$ where x is a variable and $a_n \in \mathbb{R}$ for all n . Such a series is called a **power series**.

Question: For what values of $x \in \mathbb{R}$ does the power series $\sum_{n=0}^{\infty} a_n x^n$ converge to a real number?

Clearly, any power series $\sum_{n=0}^{\infty} a_n x^n$ converges at $x = 0$ to the value a_0 . In general, the convergence of a power series is determined by the coefficients a_n . Consider the set:

$$S = \left\{ x \in \mathbb{R} \mid \sum_{n=0}^{\infty} a_n x^n \text{ converges to a real number} \right\}$$

Then S has to be one of the following subsets of $\mathbb{R}$:

1. $S = \mathbb{R}$ (Converges everywhere)

2. $S = \{0\}$ (Converges only at the center)
3. S is a bounded interval centered at 0 which is open, half-open, or closed. i.e., $S = (-\alpha, \alpha), [-\alpha, \alpha), (-\alpha, \alpha],$ or $[-\alpha, \alpha]$ for some $\alpha \in \mathbb{R}_{>0}$.

58. The Radius of Convergence

Theorem : Radius of Convergence

Let $\sum_{n=0}^{\infty} a_n x^n$ be a power series ($a_n \in \mathbb{R} \forall n$). Let $\beta = \limsup_{n \rightarrow \infty} |a_n|^{1/n}$ and $R = \frac{1}{\beta}$. (Here $R = +\infty$ if $\beta = 0$, and $R = 0$ if $\beta = +\infty$. Note $\beta \in [0, \infty]$).

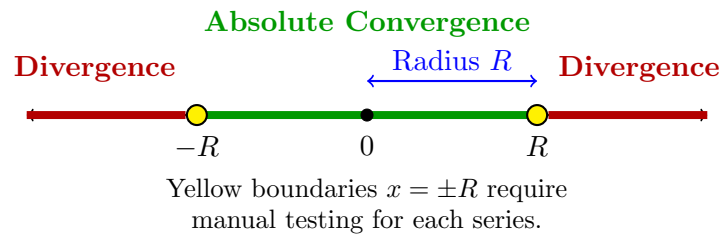
Then:

- (i) $\sum_{n=0}^{\infty} a_n x^n$ **converges absolutely** for $x \in (-R, R)$.
- (ii) $\sum_{n=0}^{\infty} a_n x^n$ **diverges** for $|x| > R$.
- (iii) $\sum_{n=0}^{\infty} a_n x^n$ may converge or diverge at the two boundary points $x = R$ and $x = -R$.

Definition : Radius of Convergence

R is called the **radius of convergence** of the power series.

Note: If $R = 0$, then (i) is a vacuous statement. If $R = \infty$, then (ii) is a vacuous statement.



Proof:

The Root Test states that if $\sum b_n$ is a series, consider $\gamma = \limsup_{n \rightarrow \infty} |b_n|^{1/n} \in [0, +\infty]$. If $\gamma < 1$, $\sum b_n$ converges absolutely. If $\gamma > 1$, $\sum b_n$ diverges.

For any $x \in \mathbb{R}$, consider the series $\sum a_n x^n$. Putting $b_n = a_n x^n$, we consider:

$$\gamma = \limsup_{n \rightarrow \infty} |b_n|^{1/n} = \limsup_{n \rightarrow \infty} |a_n x^n|^{1/n} = |x| \limsup_{n \rightarrow \infty} |a_n|^{1/n} = |x| \beta = \frac{|x|}{R}$$

Now, $\gamma < 1 \iff \frac{|x|}{R} < 1 \iff |x| < R$.

And $\gamma > 1 \iff \frac{|x|}{R} > 1 \iff |x| > R$.

Hence, if $|x| < R$, the series converges absolutely. If $|x| > R$, the series diverges.

Edge cases: If $\beta = 0$ ($R = +\infty$), then $\gamma = 0 < 1$ for all x , so it converges everywhere. If $\beta = +\infty$ ($R = 0$), $\gamma = +\infty > 1$ for all $x \neq 0$, so it converges only at $x = 0$. ■

59. Calculating R via the Ratio Test

Recall we proved that if (a_n) is a sequence of nonzero real numbers, then:

$$\liminf \left| \frac{a_{n+1}}{a_n} \right| \leq \liminf |a_n|^{1/n} \leq \limsup |a_n|^{1/n} \leq \limsup \left| \frac{a_{n+1}}{a_n} \right|$$

If the limit of the ratio exists, then all these quantities squeeze together and equal the same limit. Hence:

$$\text{If } \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \text{ exists, then } \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \beta = \limsup_{n \rightarrow \infty} |a_n|^{1/n}$$

Calculating the limit of the ratio is often much easier than calculating the lim sup of the roots.

60. Power Series Examples

Example 1: $\sum_{n=0}^{\infty} \frac{x^n}{n!}$

$a_n = \frac{1}{n!}$. Using the ratio limit:

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{n!}{(n+1)!} = \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0$$

$$\implies \beta = 0 \implies R = +\infty.$$

The series converges for all $x \in \mathbb{R}$. The interval of convergence is $(-\infty, \infty)$. (*This is the series for e^x .*)

Example 2: $\sum_{n=0}^{\infty} x^n$

$$a_n = 1. \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1 \implies \beta = 1 \implies R = 1.$$

Converges for $x \in (-1, 1)$. Diverges for $|x| > 1$.

At $x = 1$, $\sum 1^n$ diverges. At $x = -1$, $\sum (-1)^n$ diverges.

The interval of convergence is $(-1, 1)$.

Example 3: $\sum_{n=1}^{\infty} \frac{x^n}{n}$

$$a_n = \frac{1}{n}. \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{n}{n+1} = 1 \implies \beta = 1 \implies R = 1.$$

At $x = 1$: $\sum \frac{1}{n}$ diverges (Harmonic series).

At $x = -1$: $\sum \frac{(-1)^n}{n}$ converges (Alternating Series Test).

The interval of convergence is $[-1, 1)$.

Example 4: $\sum_{n=1}^{\infty} \frac{x^n}{n^2}$

$$a_n = \frac{1}{n^2}. \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{n^2}{(n+1)^2} = 1 \implies R = 1.$$

At $x = 1$: $\sum \frac{1}{n^2}$ converges (p -series with $p = 2 > 1$).

At $x = -1$: $\sum \frac{(-1)^n}{n^2}$ converges absolutely.

The interval of convergence is $[-1, 1]$.

$$\sum \frac{x^n}{n} \quad \leftarrow \bullet \text{---} \circ \rightarrow$$

-1 1

$$\sum \frac{x^n}{n^2} \quad \leftarrow \bullet \text{---} \bullet \rightarrow$$

-1 1

Both have $R = 1$, but the denominator determines if the boundary points are included.

Example 5: $\sum_{n=0}^{\infty} n!x^n$

$$a_n = n!. \quad \lim \left| \frac{a_{n+1}}{a_n} \right| = \lim \frac{(n+1)!}{n!} = \lim(n+1) = +\infty.$$

$$\implies \beta = \infty \implies R = 0.$$

The interval of convergence is $\{0\}$.

61. Shifted and Substituted Power Series

Example 6: Scaling the Exponent

Evaluate the interval of convergence for $\sum_{n=0}^{\infty} \frac{x^{3n}}{2^n} = 1 + \frac{x^3}{2} + \frac{x^6}{4} + \frac{x^9}{8} + \dots$.
Let $y = x^3$. The series becomes $\sum_{n=0}^{\infty} \frac{y^n}{2^n}$. For this new series, $a_n = \frac{1}{2^n}$.

$$\beta = \limsup_{n \rightarrow \infty} \left| \frac{1}{2^n} \right|^{1/n} = \frac{1}{2} \implies R_y = 2$$

The series converges for $|y| < 2$. Substituting back $y = x^3$:

$$|x^3| < 2 \iff |x| < 2^{1/3}$$

Thus, the radius of convergence for x is $R_x = 2^{1/3}$.

At $x = 2^{1/3}$, the series becomes $\sum \frac{(2^{1/3})^{3n}}{2^n} = \sum \frac{2^n}{2^n} = \sum 1$, which diverges.

At $x = -2^{1/3}$, the series becomes $\sum \frac{(-2^{1/3})^{3n}}{2^n} = \sum \frac{(-2)^n}{2^n} = \sum (-1)^n$, which diverges.

The interval of convergence is $(-2^{1/3}, 2^{1/3})$.

If we consider a power series of the form $\sum_{n=0}^{\infty} a_n(x - x_0)^n$, then the interval of convergence for this series will be an interval centered at x_0 . Let $y = x - x_0$. Then $\sum a_n(x - x_0)^n = \sum a_n y^n$, and the condition $|y| < R$ becomes $|x - x_0| < R$.

Example 7: Shifting the Center

Evaluate the interval of convergence for $\sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{n} (x - 1)^n$.

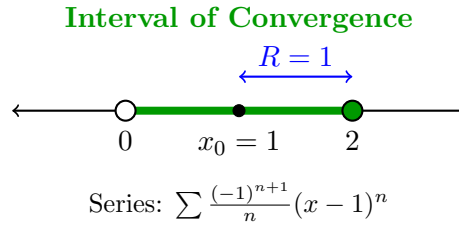
$$\text{Let } a_n = \frac{(-1)^{n+1}}{n}.$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{n}{n+1} \right| = 1 \implies \beta = 1 \implies R = 1$$

Hence this series converges for $|x - 1| < 1$, which means $x \in (0, 2)$.

At $x = 2$: The series becomes $\sum \frac{(-1)^{n+1}}{n}(1)^n = \sum \frac{(-1)^{n+1}}{n}$, which converges (Alternating Series).

At $x = 0$: The series becomes $\sum \frac{(-1)^{n+1}}{n}(-1)^n = \sum \frac{-1}{n}$, which diverges (Harmonic).
The interval of convergence is $(0, 2]$.



62. Pointwise vs. Uniform Convergence

Let $f_n : S \rightarrow \mathbb{R}$ be a sequence of functions, where $S \subseteq \mathbb{R}$ and $n \in \mathbb{N}$.

Definition : Pointwise Convergence

The sequence (f_n) **converges pointwise** to a function $f : S \rightarrow \mathbb{R}$ if:

$$\lim_{n \rightarrow \infty} f_n(x) = f(x) \quad \forall x \in S$$

Notation: $f_n \rightarrow f$ pointwise on S .

This means: $\forall \epsilon > 0, \forall x \in S, \exists N \in \mathbb{N}$ such that $|f_n(x) - f(x)| < \epsilon$ for all $n \geq N$. (Note that N depends on both ϵ and x).

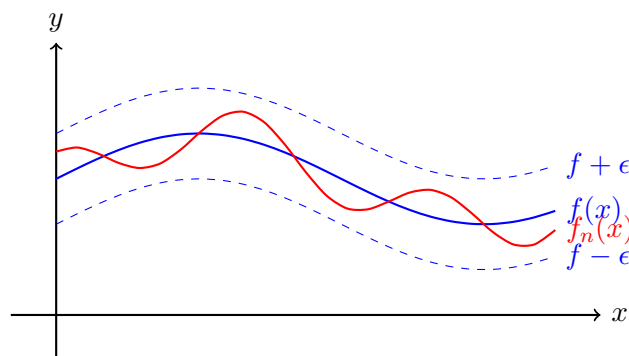
Definition : Uniform Convergence

The sequence (f_n) **converges uniformly** to $f : S \rightarrow \mathbb{R}$ if:

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } |f_n(x) - f(x)| < \epsilon \quad \forall n \geq N \text{ and } \forall x \in S$$

(Here, a single N works simultaneously for ALL $x \in S$).

i.e., the graph of f_n for $n \geq N$ lies entirely in the strip between the graphs of $f - \epsilon$ and $f + \epsilon$.



For $n \geq N$, the entire graph of f_n is trapped within the ϵ -strip.

63. The Uniform Limit of Continuous Functions

Theorem : Continuity Theorem

Let $f_n : S \rightarrow \mathbb{R}$ be a sequence of functions, $S \subseteq \mathbb{R}$. Suppose $f_n \rightarrow f$ uniformly on S . If each f_n is continuous at some point $x_0 \in S$, then f is also continuous at x_0 .

Corollary: If a sequence of continuous functions converges uniformly, the limit function f MUST be continuous. (*"The uniform limit of continuous functions is continuous"*).

Proof:

Let $\epsilon > 0$. Since $f_n \rightarrow f$ uniformly, there exists $N \in \mathbb{N}$ such that for all $n \geq N$:

$$|f_n(x) - f(x)| < \frac{\epsilon}{3} \quad \forall x \in S$$

In particular, $|f_N(x) - f(x)| < \frac{\epsilon}{3}$ for all $x \in S$.

Now, f_N is continuous at x_0 , hence there exists $\delta > 0$ such that if $x \in S$ and $|x - x_0| < \delta$, then:

$$|f_N(x) - f_N(x_0)| < \frac{\epsilon}{3}$$

Then for any $x \in S$ with $|x - x_0| < \delta$, we use the Triangle Inequality:

$$\begin{aligned} |f(x) - f(x_0)| &\leq |f(x) - f_N(x)| + |f_N(x) - f_N(x_0)| + |f_N(x_0) - f(x_0)| \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon \end{aligned}$$

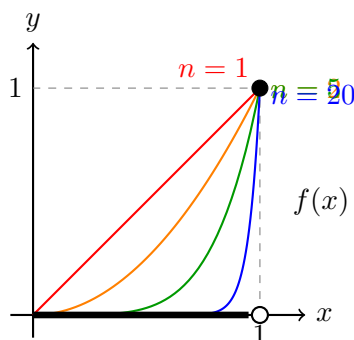
Hence, f is continuous at x_0 . ■

Example 1: Discontinuous Limit $\implies$ Not Uniform

Let $f_n : [0, 1] \rightarrow \mathbb{R}$ be defined by $f_n(x) = x^n$. The pointwise limit $f : [0, 1] \rightarrow \mathbb{R}$ is defined by:

$$f(x) = \begin{cases} 0 & \text{if } x \in [0, 1) \\ 1 & \text{if } x = 1 \end{cases}$$

Since f_n are continuous polynomials, but the limit f is discontinuous at $x = 1$, f_n **does not converge uniformly** to f .



64. Counterexamples in Uniform Convergence

Even if the limit function IS continuous, uniform convergence can still fail.

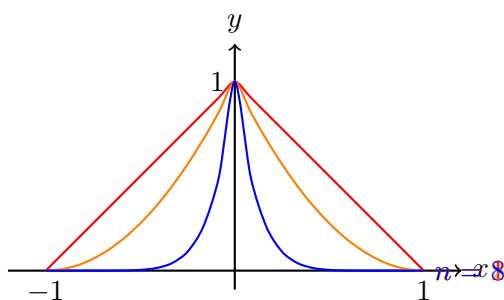
Example 2: Escaping the Epsilon Tube

Let $f_n(x) = (1 - |x|)^n$ for $x \in (-1, 1)$. The pointwise limit is:

$$f(x) = \begin{cases} 0 & \text{if } x \in (-1, 1), x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$$

Since f is not continuous at $x = 0$, convergence is not uniform.

Direct Proof: Let $\epsilon = 1/2$. Suppose $f_n \rightarrow f$ uniformly. Then $\exists N \in \mathbb{N}$ s.t. $|f(x) - f_n(x)| < 1/2$ for all x and $n \geq N$. For $x \in (0, 1)$, $f(x) = 0$, so we must have $(1-x)^N < 1/2$ for all $x \in (0, 1)$. But suppose we pick $x = 1 - \frac{1}{2^{1/N}} \in (0, 1)$. Then $(1-x)^N = \left(\frac{1}{2^{1/N}}\right)^N = \frac{1}{2}$. This contradicts $(1-x)^N < 1/2$. Thus, not uniform.



Example 3: $f_n(x) = nx^n$ on $[0, 1)$

Let $f_n : [0, 1) \rightarrow \mathbb{R}$ be $f_n(x) = nx^n$. $f_n \rightarrow 0$ pointwise on $[0, 1)$. (Proved via ratio test: $a_{n+1}/a_n = \frac{n+1}{n}x \rightarrow x < 1$).

But f_n does **not** converge to 0 uniformly. If it did, for $\epsilon = 1$, $\exists N$ s.t. $nx^n < 1$ for all $x \in [0, 1)$ and $n \geq N$. In particular, $Nx^N < 1$ for all $x \in [0, 1)$. If we pick $x = \frac{1}{N^{1/N}} \in [0, 1)$, then $x^N = \frac{1}{N}$, so $Nx^N = 1$. Contradiction!

65. The Supremum Test for Uniform Convergence

Theorem : The Supremum Test

$f_n \rightarrow f$ uniformly on $S \iff \lim_{n \rightarrow \infty} \sup\{|f_n(x) - f(x)| \mid x \in S\} = 0$.

If $f - f_n$ is differentiable, we can use standard Calculus optimization (finding the derivative, setting it to 0) to find this supremum!

Example 4: $\frac{1}{n} \sin(nx)$ (Uniform)

$f_n(x) = \frac{1}{n} \sin(nx)$ on $\mathbb{R}$. Pointwise limit is $f(x) = 0$.

$$\sup_{x \in \mathbb{R}} \left| \frac{1}{n} \sin(nx) - 0 \right| \leq \frac{1}{n}$$

Since $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$, $f_n \rightarrow 0$ uniformly on $\mathbb{R}$.

Example 5: $f_n(x) = \frac{x}{1+nx^2}$ (Uniform)

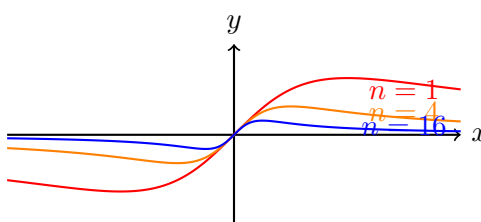
Pointwise limit is $f(x) = 0$. We want to find the max of f_n on $\mathbb{R}$.

$$f'_n(x) = \frac{(1+nx^2) - x(2nx)}{(1+nx^2)^2} = \frac{1-nx^2}{(1+nx^2)^2} = 0 \implies x = \pm \frac{1}{\sqrt{n}}$$

Evaluating at the maximum: $f_n\left(\frac{1}{\sqrt{n}}\right) = \frac{1/\sqrt{n}}{1+1} = \frac{1}{2\sqrt{n}}$.

$$\lim_{n \rightarrow \infty} \sup_{x \in \mathbb{R}} |f_n(x)| = \lim_{n \rightarrow \infty} \frac{1}{2\sqrt{n}} = 0$$

Thus, $f_n \rightarrow 0$ **uniformly** on $\mathbb{R}$.



The peaks occur at $x = 1/\sqrt{n}$ and their heights shrink to 0. Thus, convergence is uniform.

Example 6: $f_n(x) = n^2 x^n (1-x)$ (Not Uniform)

Pointwise limit is $f(x) = 0$ on $[0, 1]$. We find the maximum:

$$f'_n(x) = n^2 x^{n-1} (n(1-x) - x) = 0 \implies x = 0 \text{ or } x = \frac{n}{n+1}$$

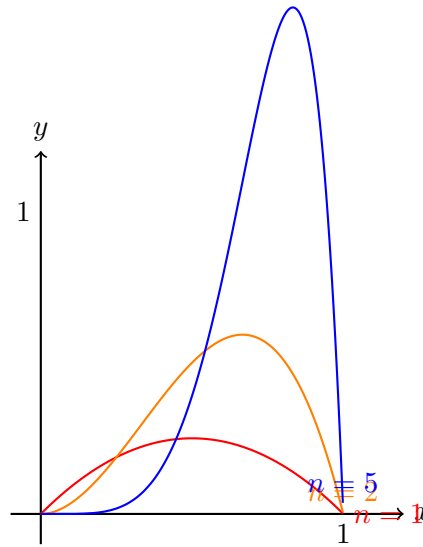
Evaluating at the maximum:

$$f_n\left(\frac{n}{n+1}\right) = n^2 \left(\frac{n}{n+1}\right)^n \left(1 - \frac{n}{n+1}\right) = \frac{n^2}{n+1} \left(\frac{n}{n+1}\right)^n$$

We know $\lim_{n \rightarrow \infty} \left(\frac{n}{n+1}\right)^n = \frac{1}{e}$.

$$\lim_{n \rightarrow \infty} f_n\left(\frac{n}{n+1}\right) = \lim_{n \rightarrow \infty} \frac{n^2}{n+1} \left(\frac{1}{e}\right) = \infty$$

Since the supremum goes to ∞ , f_n **does not** converge uniformly.



The peak moves right and grows to ∞ .
It escapes any ϵ -tube!

66. Series of Functions

We now consider infinite series whose terms are functions rather than constants.

Definition : Convergence of a Series of Functions

Let (f_k) be a sequence of functions defined on $S \subseteq \mathbb{R}$. We define the n -th partial sum function $g_n : S \rightarrow \mathbb{R}$ by:

$$g_n(x) = \sum_{k=1}^n f_k(x) = f_1(x) + f_2(x) + \cdots + f_n(x)$$

The series of functions $\sum_{k=1}^{\infty} f_k(x)$ converges **pointwise** to a function $g(x)$ on S if the sequence of partial sums (g_n) converges pointwise to g on S .

The series converges **uniformly** to $g(x)$ on S if the sequence of partial sums (g_n) converges uniformly to g on S .

Theorem : Cauchy Criterion for Uniform Convergence of Series

The series $\sum_{k=1}^{\infty} f_k(x)$ converges uniformly on S if and only if for every $\epsilon > 0$, there exists an integer N such that for all $n > m \geq N$:

$$\sup_{x \in S} \left| \sum_{k=m+1}^n f_k(x) \right| = \sup_{x \in S} |f_{m+1}(x) + \cdots + f_n(x)| < \epsilon$$

Corollary (Uniform n -th Term Test): If a series $\sum f_k(x)$ converges uniformly on S , then the sequence of individual terms $f_k(x)$ must converge uniformly to 0 on S .

67. The Weierstrass M-Test

The most powerful tool for proving uniform convergence of a series bypasses the Cauchy criterion by relying on a dominating series of constants.

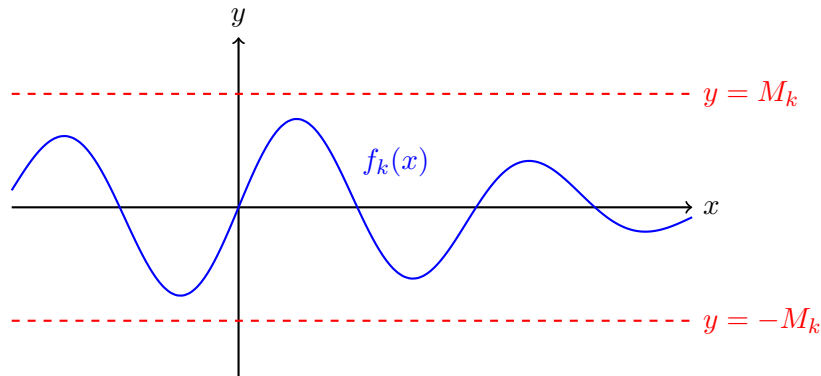
Theorem : Weierstrass M-Test

Let (f_k) be a sequence of functions on $S \subseteq \mathbb{R}$. Suppose there exists a sequence of real constants $M_k \geq 0$ such that:

$$|f_k(x)| \leq M_k \quad \text{for all } x \in S \text{ and for all } k$$

(Equivalently, $\sup_{x \in S} |f_k(x)| \leq M_k$).

If the constant series $\sum_{k=1}^{\infty} M_k$ **converges**, then the function series $\sum_{k=1}^{\infty} f_k(x)$ converges **uniformly and absolutely** on S .



The entire function graph is permanently trapped beneath the constant ceiling M_k .

Proof:

We use the Cauchy Criterion. Since $\sum M_k$ converges, it satisfies the Cauchy condition for real numbers. Given $\epsilon > 0$, there exists N such that for all $n > m \geq N$, $|M_{m+1} + \cdots + M_n| < \epsilon$. Because $M_k \geq 0$, this means $M_{m+1} + \cdots + M_n < \epsilon$.

For any $x \in S$, using the Triangle Inequality:

$$\left| \sum_{k=m+1}^n f_k(x) \right| \leq \sum_{k=m+1}^n |f_k(x)| \leq \sum_{k=m+1}^n M_k < \epsilon$$

Since this strict bound holds for all $x \in S$ simultaneously, the supremum of the partial sum tail is less than ϵ . By the Cauchy Criterion for Uniform Convergence, the series converges uniformly. (Because we bounded the absolute values, it also converges absolutely). ■

68. Applications of the M-Test

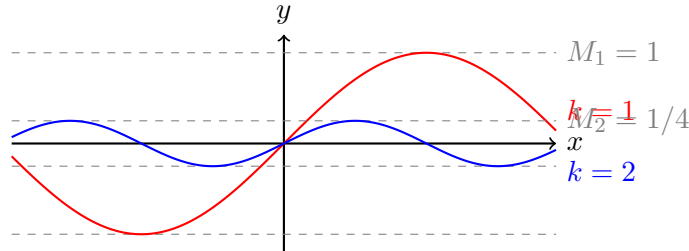
Example 1: Fourier-style Series on $\mathbb{R}$

Show that $\sum_{k=1}^{\infty} \frac{\sin(kx)}{k^2}$ converges uniformly on $\mathbb{R}$.

Solution: Let $f_k(x) = \frac{\sin(kx)}{k^2}$. Since $|\sin(\theta)| \leq 1$ for all real θ , we have:

$$|f_k(x)| = \left| \frac{\sin(kx)}{k^2} \right| \leq \frac{1}{k^2} \quad \forall x \in \mathbb{R}$$

Let $M_k = \frac{1}{k^2}$. The series $\sum_{k=1}^{\infty} M_k = \sum_{k=1}^{\infty} \frac{1}{k^2}$ is a convergent p -series ($p = 2 > 1$). By the Weierstrass M-Test, the series of functions converges uniformly and absolutely on $\mathbb{R}$.



Example 2: Rational Function Series

Show that $\sum_{k=1}^{\infty} \frac{1}{x^2 + k^2}$ converges uniformly on $\mathbb{R}$.

Solution: For any $x \in \mathbb{R}$, $x^2 \geq 0$, which means $x^2 + k^2 \geq k^2$. Taking the reciprocal flips the inequality:

$$\left| \frac{1}{x^2 + k^2} \right| \leq \frac{1}{k^2} \quad \forall x \in \mathbb{R}$$

Again, $M_k = \frac{1}{k^2}$, and $\sum M_k$ converges. By the M-Test, the series converges uniformly on $\mathbb{R}$.

Example 3: Where the M-Test Fails

Analyze the convergence of $\sum_{k=1}^{\infty} \frac{x^2}{x^2 + k^2}$ on $\mathbb{R}$.

Solution: Let $f_k(x) = \frac{x^2}{x^2 + k^2}$. If we seek a global bound M_k , we must find the supremum

of f_k over $\mathbb{R}$.

$$\lim_{x \rightarrow \infty} \frac{x^2}{x^2 + k^2} = 1$$

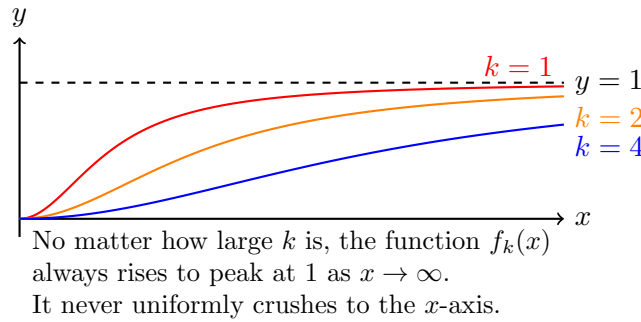
Thus, $\sup_{x \in \mathbb{R}} |f_k(x)| = 1$. The tightest bound we can use is $M_k = 1$. However, $\sum_{k=1}^{\infty} 1 = \infty$. The M-Test fails on $\mathbb{R}$.

Does it still converge uniformly, just without the M-test? **No.** By the Uniform n -th Term Test, if a series converges uniformly, the individual terms must converge uniformly to 0. But $\sup |f_k(x)| = 1 \neq 0$. Thus, $f_k(x)$ does not converge uniformly to 0. The series **does not** converge uniformly on $\mathbb{R}$.

Restriction to a bounded interval: If we restrict $x \in [-a, a]$ for some fixed $a > 0$:

$$\left| \frac{x^2}{x^2 + k^2} \right| \leq \frac{a^2}{0 + k^2} = \frac{a^2}{k^2}$$

Let $M_k = \frac{a^2}{k^2} = a^2 \left(\frac{1}{k^2} \right)$. Since $\sum \frac{a^2}{k^2}$ converges, the series **does** converge uniformly on any bounded interval $[-a, a]$.



69. Integration of Sequences and Series

The fundamental question is whether we can swap the order of a limit and an integral:

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx \stackrel{?}{=} \int_a^b \left(\lim_{n \rightarrow \infty} f_n(x) \right) dx$$

Pointwise convergence is insufficient to guarantee this equality. Uniform convergence is required.

Theorem : Uniform Convergence and Integration

Let $f_n : [a, b] \rightarrow \mathbb{R}$ be a sequence of continuous functions. If $f_n \rightarrow f$ **uniformly** on $[a, b]$, then f is continuous (and therefore integrable) and:

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b f(x) dx$$

Proof:

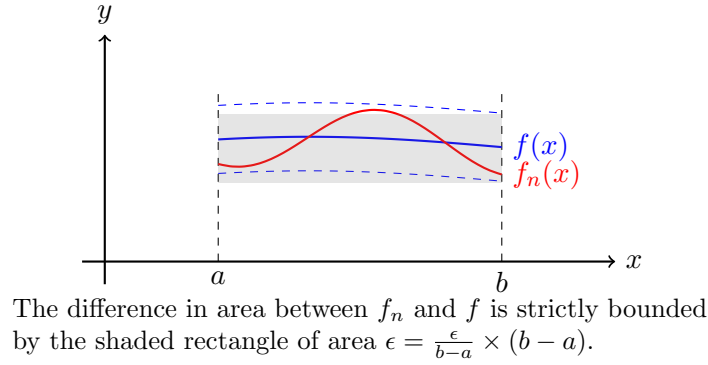
Let $\epsilon > 0$. Because $f_n \rightarrow f$ uniformly, there exists $N \in \mathbb{N}$ such that for all $n \geq N$:

$$|f_n(x) - f(x)| < \frac{\epsilon}{b-a} \quad \text{for all } x \in [a, b]$$

We want to show that the difference between the integrals goes to 0. Using properties of integrals:

$$\begin{aligned} \left| \int_a^b f_n(x) dx - \int_a^b f(x) dx \right| &= \left| \int_a^b (f_n(x) - f(x)) dx \right| \\ &\leq \int_a^b |f_n(x) - f(x)| dx \\ &< \int_a^b \frac{\epsilon}{b-a} dx = \frac{\epsilon}{b-a} (b-a) = \epsilon \end{aligned}$$

Thus, for $n \geq N$, the difference between the integrals is strictly less than ϵ . This proves the limit of the integrals equals the integral of the limit. ■



Theorem : Term-by-Term Integration of Series

If a series of continuous functions $\sum_{k=1}^{\infty} f_k(x)$ converges **uniformly** to $g(x)$ on $[a, b]$, then we can integrate the series term-by-term:

$$\int_a^b \sum_{k=1}^{\infty} f_k(x) dx = \sum_{k=1}^{\infty} \int_a^b f_k(x) dx = \int_a^b g(x) dx$$

70. Differentiation of Sequences and Series

Derivatives are sensitive to high-frequency oscillations. Uniform convergence of $f_n \rightarrow f$ is **not enough** to guarantee that $f'_n \rightarrow f'$.

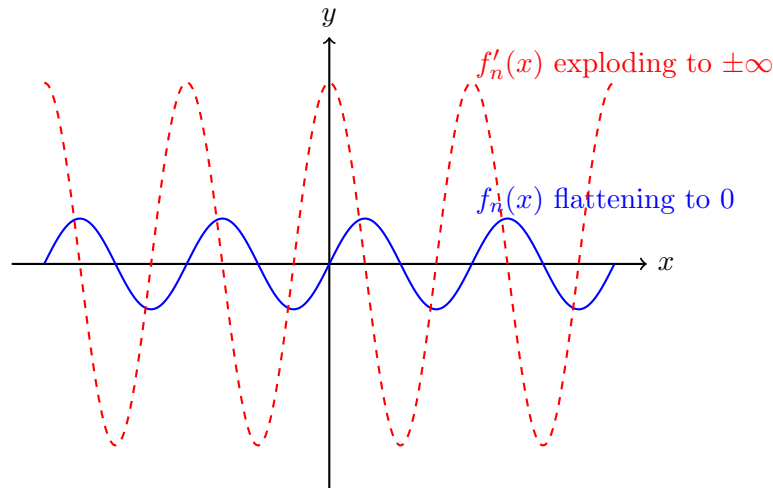
The Classic Counterexample

Let $f_n(x) = \frac{\sin(nx)}{\sqrt{n}}$ for $x \in \mathbb{R}$. Because $\left| \frac{\sin(nx)}{\sqrt{n}} \right| \leq \frac{1}{\sqrt{n}}$, and $\frac{1}{\sqrt{n}} \rightarrow 0$, $f_n \rightarrow 0$ **uniformly** on $\mathbb{R}$. The limit function is $f(x) = 0$, so $f'(x) = 0$.

However, look at the derivatives:

$$f'_n(x) = \frac{n \cos(nx)}{\sqrt{n}} = \sqrt{n} \cos(nx)$$

Evaluating at $x = 0$, $f'_n(0) = \sqrt{n} \cos(0) = \sqrt{n}$. As $n \rightarrow \infty$, $f'_n(0) \rightarrow \infty \neq f'(0)$. The derivatives do not converge!



Even as the blue wave flattens perfectly to the x-axis, its slopes (the red wave) become infinitely steep.

Theorem : Uniform Convergence and Differentiation

Let f_n be a sequence of differentiable functions on $[a, b]$. Suppose that:

1. The sequence of **derivatives** f'_n converges **uniformly** to a function g on $[a, b]$.
2. The original sequence $f_n(x_0)$ converges pointwise for at least one point $x_0 \in [a, b]$.

Then f_n converges uniformly on $[a, b]$ to a function f , and f is differentiable, and $f'(x) = g(x)$. (For series: if the series of derivatives converges uniformly, we can differentiate term-by-term).

71. Calculus of Power Series

Power series $\sum a_n x^n$ are perfectly behaved inside their radius of convergence R .

Theorem : Term-by-Term Operations on Power Series

Let $\sum a_n x^n$ be a power series with radius of convergence $R > 0$. Then, for any $x \in (-R, R)$:

1. The series can be differentiated term-by-term:

$$\frac{d}{dx} \left(\sum_{n=0}^{\infty} a_n x^n \right) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

2. The series can be integrated term-by-term:

$$\int_0^x \left(\sum_{n=0}^{\infty} a_n t^n \right) dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$$

3. Both the differentiated series and the integrated series have the **exact same radius of convergence** R .

Proof:

Let's prove the radius remains R for the derivative. We use the Root Test on the new coefficients na_n :

$$\limsup_{n \rightarrow \infty} |na_n|^{1/n} = \limsup_{n \rightarrow \infty} \left(n^{1/n} |a_n|^{1/n} \right)$$

It is a known limit that $\lim_{n \rightarrow \infty} n^{1/n} = 1$. Therefore:

$$\limsup_{n \rightarrow \infty} \left(n^{1/n} |a_n|^{1/n} \right) = 1 \cdot \limsup_{n \rightarrow \infty} |a_n|^{1/n} = \beta$$

Since β is unchanged, the new radius $R' = 1/\beta = R$. The same logic applies to the integral using $\lim_{n \rightarrow \infty} \left(\frac{1}{n+1} \right)^{1/n} = 1$. ■

Application: Finding the Taylor Series for $\ln(1-x)$

We know the geometric series:

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots \quad \text{for } x \in (-1, 1)$$

Because this is a power series, we can integrate both sides term-by-term from 0 to x :

$$\begin{aligned} \int_0^x \frac{1}{1-t} dt &= \int_0^x \left(\sum_{n=0}^{\infty} t^n \right) dt = \sum_{n=0}^{\infty} \int_0^x t^n dt \\ -\ln(1-x) &= \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1} = x + \frac{x^2}{2} + \frac{x^3}{3} + \dots \end{aligned}$$

Multiplying by -1 :

$$\ln(1-x) = - \sum_{n=1}^{\infty} \frac{x^n}{n} \quad \text{for } x \in (-1, 1)$$

72. Abel's Theorem

We know a power series is continuous strictly *inside* $(-R, R)$. But what if the series successfully converges exactly at the boundary point $x = R$?

Theorem : Abel's Limit Theorem

Let $\sum a_n x^n$ be a power series with radius of convergence R . If the series converges at the boundary point $x = R$, then the function $f(x) = \sum a_n x^n$ is continuous from the left at $x = R$. That is:

$$\lim_{x \rightarrow R^-} \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} a_n R^n$$